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# Validating Causal Abstraction Metrics on Simulated Complex Systems

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## Abstract

1 A central goal of science is to produce valid explanations of complex systems:  
2 high-level causal accounts that faithfully reflect the behavior of lower-level mech-  
3 anisms. Yet no consensus exists on how to measure whether a proposed high-level  
4 explanation is actually valid. We introduce a benchmark of ten complex systems  
5 spanning both discrete and continuous state spaces, as well as static and dynamical  
6 regimes, each equipped with consensual ground-truth causal explanations and  
7 invalid contrastive conditions. Within a unified causal abstraction framework, we  
8 systematically evaluate over thirty candidate metrics drawn from observational,  
9 functional, information-theoretic, and causal families. Our results show that only  
10 the latter reliably discriminates valid from invalid abstractions, and only when incor-  
11 porating faithfulness testing over unmapped variables. Building on these findings,  
12 we introduce the Causal Abstraction Error (CAE), a continuous validity metric  
13 with an explicit faithfulness test, which passes all discrimination tests across every  
14 system and converges with as few as 30 sampled interventions. We offer it as a  
15 general-purpose metric for the discovery and validation of high-level explanations.

## 16 1 Introduction

17 A central goal of science is to produce **explanations**: not merely descriptions or predictions, but  
18 accounts of *why* phenomena occur [Hempel, 1965, Woodward, 2004]. While the philosophy of  
19 science has long debated what explanations are, a working consensus has emerged around a few  
20 core desiderata: good scientific explanations should be causally informative [Woodward, 2004, Pearl,  
21 2009], parsimonious [Kitcher, 1962, Salmon, 1984, Batterman and Rice, 2014], and appropriately  
22 scoped to the context and level of description at which the target phenomenon is best characterized  
23 [Lombrozo, 2006, Potochnik, 2017]. Finding good explanations is particularly difficult when the  
24 system under study exhibits what Warren Weaver famously called *organized complexity* [Weaver,  
25 1991]: systems with many interacting parts that are neither so disordered as to yield to statistical  
26 averaging, nor so simple as to admit direct analytic treatment. The appropriate high-level variables  
27 for explanation must be discovered, and the class of functions that legitimately aggregate low-level  
28 quantities into those variables is itself a subject of inquiry [Hoel et al., 2013, Potochnik, 2017].

29 This raises a fundamental challenge: what are useful high-level variables, and how should they be  
30 defined from lower-level quantities? Different fields studying different classes of complex systems  
31 have developed their own candidate answers, such as firing rates and population codes in systems  
32 neuroscience [Cunningham and Yu, 2014], or species abundance and trophic levels in ecology [Loreau,  
33 2010]. However, a unified cross-disciplinary methodology for evaluating high-level explanations of  
34 complex systems remains elusive. A sobering illustration of why comes from Jonas and Kording  
35 [2017], who applied the standard causal and statistical toolkit of neuroscience to a fully observable  
36 microprocessor: a relatively simple system engineered for modular, hierarchical organization. In  
37 doing so, they failed to recover meaningful high-level properties.

The recent success of artificial intelligence seems to offer a way forward. Modern AI systems are themselves structurally complex computational systems, yet they are fully observable and perfectly manipulable, a rare property in the natural sciences. This privileged epistemic access makes AI systems a methodological laboratory for the science of complex systems [Holtzman et al., 2025], to develop and test explanation discovery procedure in simplified settings. Yet, AI interpretability has also proven sobering, with a large tendency to produce false positive findings and explanations that are unstable and not generalizable [Méloux et al., 2025a].

A general metric for evaluating proposed high-level explanations of low-level complex systems is therefore a prerequisite to move forward: without it, the discovery and evaluation of candidate cannot be done rigorously, and progress is hard to measure or transfer across domains. Yet, no such agreed-upon, general metric exists. A growing body of work has begun addressing this gap through the lens of **causal abstraction** [Beckers and Halpern, 2019], the idea that a high-level causal model is a valid explanation of a low-level system if interventions at the high level correspond faithfully to interventions at the low level under a postulated abstraction map. This framework is conceptually appealing and promising, but existing implementations differ in important ways: in whether they treat abstraction as exact or approximate [Beckers et al., 2020], and in how they handle distributions over inputs or over interventions [Geiger et al., 2021, 2025]. Current concrete metric implementations developed in the field of AI interpretability have also been shown to suffer from the global issues of striking false positives and non-generalizable findings [Sutter et al., 2025, Méloux et al., 2025a].

**In this work**, we set out to systematically test, compare, and benchmark a wide range of proposed explanation metrics, observational, information-theoretic, symbolic, and causal, across a diverse collection of idealized complex systems with known, consensual high-level explanations. Rather than confining to a single domain, we deliberately span systems that vary across multiple dimensions: (i) with discrete versus continuous state spaces, (ii) with static versus temporal dynamics, (iii) with disordered versus structured complexity (in Weaver’s sense). For each system, we manually construct the alignment function mapping low-level variables to high-level ones, implement all pairs within a shared conceptual formalism (causal abstraction) and a unified programmatic API supporting both observational and interventional queries. Finally, we handcraft contrastive perturbations, modifications to either the low-level or high-level model that render the proposed explanation invalid, allowing us to test whether each candidate metric correctly recognizes broken explanations as well as valid ones. This is an empirical test of the practical operationalization of existing conceptual ideas.

Our results provide strong support for the causal abstraction research program, but only for a specific distributional variants. We further introduce the Causal Abstraction Error (CAE), an error measure that incorporates a principled treatment of distributional inputs and interventions, a built-in faithfulness measurement, and produces a real-valued degree-of-violation score. The CAE passes all contrastive tests across every system class we examine and outperforms related causal abstraction variants.

## 2 Metrics of Explanation Validity

Following the framework of Pearl [2009] and Beckers and Halpern [2019], we model the system under study as a *structural causal model* (SCM) to implement the semantics of interventions and counterfactuals. Dyer et al. [2024] previously established that simulated complex systems belong to this class.

**Definition 2.1 (Structural Causal Model)** A *structural causal model* is a tuple  $\mathcal{M} = \langle \mathcal{U}, \mathcal{V}, \mathcal{R}, \mathcal{F}, P_{\mathcal{U}} \rangle$ , where  $\mathcal{U}$  is a set of exogenous variables with joint distribution  $P_{\mathcal{U}}$ ;  $\mathcal{V}$  is a set of endogenous variables with ranges  $\mathcal{R} = \{\mathcal{R}_V\}_{V \in \mathcal{V}}$ ; and  $\mathcal{F} = \{f_V\}_{V \in \mathcal{V}}$  is a set of structural equations, each determining  $V$  from its direct causes  $\text{Pa}(V) \subseteq \mathcal{V}$  and a noise term  $\mathcal{U}_V \subseteq \mathcal{U}$ .

A *computational explanation* is a surrogate model  $\mathcal{E}$  taken from some admissible class  $\mathfrak{E}$ , intended to account for the behavior of  $\mathcal{M}$  at a relevant level of abstraction. The central question studied in this paper is: *Under what conditions is  $\mathcal{E}$  a valid computational explanation of  $\mathcal{M}$  under  $P_{\mathcal{U}}$ ?* Radically different answers to this question have been proposed. We briefly discuss here the ones we evaluate and offer a more in-depth description in Appendix B.

**Observational validity.** The most elementary criterion is *observational equivalence*:  $\mathcal{E}$  should reproduce the measurable outputs of  $\mathcal{M}$  under the *natural* input distribution. For deterministic settings, this is operationalized via pointwise discrepancy metrics, including MSE, RMSE,  $L^2$ , and their nor-

malized variant NMSE, together with the coefficient of determination  $R^2$  [Koza, 1994, Ljung, 1999]. Complexity-regularized variants such as AIC [Akaike, 2003], BIC [Schwarz, 1978], MDL [Mangan et al., 2017], and Mallows’s  $C_p$  [Lee and Ghosh, 2009] penalize explanation complexity to prevent overfitting. When  $\mathcal{M}$  is stochastic, distributional similarity is measured via measures like KL divergence, the symmetric JS divergence, or the kernel-based MMD [Gretton et al., 2012]; HSIC [Gretton et al., 2005] can further tests whether residuals of the explanation are independent of the input, probing global agreement. For dynamical systems, validity must hold over entire trajectories: we evaluate trajectory MSE, dynamic time warping (DTW; Sakoe and Chiba, 1978), temporal autocorrelation matching, and spectral analysis [Percival and Walden, 1993]. Finally, we also consider symbolic regression methods like the SINDy evaluator [Brunton et al., 2016], which scores an explanation by the residual of its governing equations on the observed state derivatives. All observational criteria share a fundamental limitation called *equivfinality* [Valogianni and Padmanabhan, 2022, Collins et al., 2024]: distinct mechanisms can be observationally indistinguishable, yielding explanations that often do not generalize [Ghorbani et al., 2019, Kindermans et al., 2019, Méloux et al., 2025b].

**Functional validity.** Functional criteria require that  $\mathcal{E}$  additionally reproduce the *input–output response profile* of  $\mathcal{M}$ . *Variance decomposition* via ANOVA or Sobol sensitivity indices [Sobol’, 2001] decomposes output variance into per-input contributions and interactions, with validity requiring agreement between the indices of the high-level model  $\mathcal{E}$  and the low-level model  $\mathcal{M}$ . At the local level, the infidelity metric [Yeh et al., 2019], adapted to the two-model setting, measures whether  $\mathcal{E}$  and  $\mathcal{M}$  agree on their per-input attribution vectors. This replaces the original single-model attribution formulation with a two-model sensitivity comparison, using  $\mathcal{E}$  itself as the local linear approximation. LIME [Ribeiro et al., 2016] and SHAP [Lundberg and Lee, 2017] can be understood as proxies for this notion. Globally, relational fidelity [Collins et al., 2024] requires consistency of output differences across all input pairs, ensuring that  $\mathcal{E}$  preserves the functional geometry of  $\mathcal{M}$ . However, functional criteria still remain agnostic to internal mechanisms.

**Information-theoretic and representational validity.** A third class of criteria requires that  $\mathcal{E}$  reproduce the representational structure of  $\mathcal{M}$ . This can be measured by representational similarity analysis [Kriegeskorte et al., 2008]. Similarly, *probing accuracy* [Alain and Bengio, 2018] measures whether the variables posited by  $\mathcal{E}$  are linearly decodable from the internal states of  $\mathcal{M}$ . The *information bottleneck* (IB) Lagrangian [Tishby et al., 2000] characterizes valid representations as those achieving an optimal compression–relevance trade-off. *Complexity shift* [Zenil et al., 2019] measures whether  $\mathcal{E}$  induces comparable transformations under perturbation, assessed via Kolmogorov complexity. These criteria capture *what* information is present, but not *how* it is causally transformed [Elazar et al., 2021, Lasri et al., 2022, Teney et al., 2022].

**Causal and interventional validity.** The strictest criteria require  $\mathcal{E}$  to correctly reproduce  $\mathcal{M}$ ’s behavior under *active interventions*. *Symbion* [Gritti et al., 2020], focused on program explanation, scores explanations by the fraction of execution steps at which symbolic and concrete states agree, providing a tractable formal-verification proxy. The *causal sensitivity index* and the related *structural deviation* metric [Katende, 2025] measure how much the interventional consistency score between  $\mathcal{E}$  and  $\mathcal{M}$  changes when each parameter of  $\mathcal{E}$  is zeroed-out or perturbed by a fixed fraction, respectively. Mechanistic interpretability [Olah et al., 2020] seeks to recover the sparse *circuits* [Vig et al., 2020, Meng et al., 2022] implementing a given behavior, with validity defined by output agreement under computational interventions. Finally,  $\epsilon$ -machines [Crutchfield and Young, 1989, Shalizi and Crutchfield, 2001] define the canonical valid explanation as the minimal sufficient statistic for predicting  $\mathcal{M}$  the next steps of a dynamical system, potentially under interventions. The *dynamic causal consistency* (DCC) criterion [Grosse-Wentrup et al., 2024] offers a practical implementation of this idea.

## 2.1 Zooming in on Causal Abstraction

Causal abstraction aims to provide a mathematical foundation for determining when high-level models can safely be considered as causal proxies for low-level ones [Chalupka et al., 2016, Rubenstein et al., 2017, Beckers and Halpern, 2019]. It is the central conceptual framework in this work. We explore its formal definitions and the ongoing debates regarding its precise framing.

**Overview.** We denote  $\mathcal{M}$  as the micro- (or low-level) model and  $\mathcal{E}$  as the macro- (or high-level) model.  $\mathcal{E}$  is now also defined as an SCM. An abstraction between these models is defined by a

mapping  $\tau : \mathcal{M} \rightarrow \mathcal{E}$  which specifies how low-level computations in  $\mathcal{M}$  translate to high-level ones in  $\mathcal{E}$ . Zennaro [2022] reviews properties of existing definitions of valid abstractions and their evolution [Rubenstein et al., 2017, Beckers and Halpern, 2019, Beckers et al., 2020, Rischel, 2020, Rischel and Weichwald, 2021, Otsuka and Saigo, 2022]. Two key categories of properties are relevant for us: **formal structural properties** of the mapping itself and **consistency properties**, which ensure that reasoning with either the micro- or the macro-model leads to causally consistent outputs.

**Structural Properties.** Structural properties define constraints on the abstraction mapping  $\tau$ , determining which types of abstraction maps are *permissible*. Minimal but sufficient properties to model the complex systems we consider are as follows:

**1. Surjective disjoint coarse-graining of variables:** The variables in the micro-model should be coarse-grained into variables of the macro-model through a surjective mapping  $a$ , called the **coarse-graining map**. This map assigns groups of micro-variables to macro-variables. If the coarse-graining is disjoint, meaning that the same micro-variables cannot compute different macro-variables, we speak of **constructive causal abstraction** [Beckers and Halpern, 2019], which is widely used [Zennaro et al., 2023a, Kekić et al., 2023, Zhu et al., 2024]. The surjectivity of  $a$  ensures that each macro-variable is derived from some micro-variables.

**2. Surjective value mapping:** The coarse-graining map only associates variables (structural). We must also specify how micro-states map to macro-states (contents). This is achieved via a set of **value maps**  $\tau_X : a^{-1}(X) \rightarrow X$ , which determines, for each macro-variable  $X$ , how its state arises from the states of its associated micro-variables. We require each value map to be surjective to guarantee that every possible macro-state can be realized from some micro-states.

Finally, the abstraction map  $\tau$  consists of both a surjective (disjoint) coarse-graining map  $a$  and the set of all surjective value maps  $(\tau_X)$ .

**Consistency Property.** For a given a macro-intervention  $do(X = x)$ , written  $do_\nu$ , multiple compatible micro-level interventions  $do(a^{-1}(X) = \tau_X^{-1}(x))$  exist, since  $\tau_X$  is not typically injective (multiple micro-states may realize the same macro-state). Let  $do_\mu$  be one such micro-intervention. Then, the consistency property requires that there must exist a noise-variable map  $\tau_u : \mathcal{U}_\mathcal{M} \rightarrow \mathcal{U}_\mathcal{E}$  such that for every compatible pair of interventions  $(\nu, \mu)$  and every  $u \in \mathcal{U}_\mathcal{M}$ ,  $\mathcal{E}(\mathcal{V}_\mathcal{E} | \tau_u(u), do_\nu) = \tau(\mathcal{M}(\mathcal{V}_\mathcal{M} | u, do_\mu))$ . This pointwise counterfactual consistency requires that both models coincide for every exogenous noise realization. When  $\mathcal{E}$  is deterministic, this equation reduces to  $\mathcal{E}(\mathcal{V}_\mathcal{E} | do_\nu) = \tau(\mathcal{M}(\mathcal{V}_\mathcal{M} | u, do_\mu))$ , which is the primary criterion implemented in this work. This typically visualized via a commuting diagram [Beckers and Halpern, 2019].

**Measuring Consistency Errors.** From the consistency requirement, it is possible to derive an error measure, quantifying how far a macro-model (and its associated abstraction map) is from being a valid causal abstraction of a given micro-model. A general formulation of such a measure is given by:

$$E_\tau(\nu, \mu) = D(\mathcal{E}(Y | do_\nu), \tau_Y(\mathcal{M}(a^{-1}(Y) | do_\mu))),$$

where  $D$  denotes a pointwise dissimilarity metric (e.g., MSE) in the deterministic setting, or a distributional divergence for stochastic model. The function  $E_\tau(\nu, \mu)$  quantifies the degree of consistency violation for a given macro-intervention  $\nu$  and one associated micro-intervention  $\mu$ . Different approaches have been proposed to define a *global* consistency error: Zennaro et al. [2023a] adopts the worst-case error over  $(\nu, \mu)$ , defining  $E_\tau = \sup E_{\nu, \mu}$  and using the Jensen-Shannon divergence as the choice for  $D$ . Then, Zhu et al. [2024] and Kekić et al. [2023] employ the Kullback-Leibler divergence and compute the expectation over all single-variable interventions, facilitating sample-based approximations. A different approach is intervention interchange accuracy (IIA; Geiger et al., 2022a), a metric developed especially for neural networks. IIA fits into the consistency framework above but checks it only for output variables, not for intermediate ones, and does not require a complete abstraction map. Consequently, even perfect alignment under IIA does not guarantee that the macro-model is a causal abstraction of the micro-model in the general sense, since intermediate variables may still violate consistency [Méloux et al., 2025a, Sutter et al., 2025].

**Faithfulness.** The standard definition of causal abstraction does not require all low-level variables in  $\mathcal{V}_\mathcal{M}$  to be used. This is natural for AI interpretability, where explanations often posit that only a small subset of a neural network is relevant to a behavior [Olah et al., 2020, Conmy et al., 2023],

and similar sparsity assumptions appear in biological systems [Rochefort et al., 2009, Meunier et al., 2010]. Yet, declaring excluded components causally irrelevant is a strong empirical claim, that must be tested. In circuit discovery, this is captured by the **faithfulness** criterion: unmapped neurons should be intervened on to verify that they do not influence the proposed circuit computation [Hanna et al., 2024]. Existing operationalizations of causal abstraction largely take the relevant variable set as modeler-provided, rather than testing whether excluded variables are genuinely inert [Zennaro et al., 2023a, Kekić et al., 2023, Zhu et al., 2024]. Enforcing faithfulness would reject explanations that incorrectly assume away causal influence from omitted components.

### 3 The Constructive Abstraction Error

Building upon existing causal alignment error measures [Zennaro et al., 2023a, Kekić et al., 2023, Zhu et al., 2024], we define a new metric that explicitly captures faithfulness in causal abstraction. Let  $\Phi$  denote the set of *phi variables*: micro-variables that are neither mapped to any macro-variable by  $a$  nor declared as causally redundant internal variables of  $\mathcal{M}$ . We define two variants of CAE that differ in the direction of sampling through the consistency diagram:

$$\text{CAE}_{\downarrow}(\mathcal{M}, \mathcal{E}, \Phi, \tau) = \mathbb{E}_{\nu \sim P_I} \mathbb{E}_{\mu \sim P_{\nu}} D(\mathcal{E}(Y \mid \text{do}_{\nu}), \tau_Y(\mathcal{M}(a^{-1}(Y) \mid \text{do}_{\mu}))), \quad (1)$$

$$\text{CAE}_{\uparrow}(\mathcal{M}, \mathcal{E}, \Phi, \tau) = \mathbb{E}_{\mu \sim P_I} D(\mathcal{E}(Y \mid \text{do}_{\tau(\mu)}), \tau_Y(\mathcal{M}(a^{-1}(Y) \mid \text{do}_{\mu}))). \quad (2)$$

where  $D$  is a pointwise dissimilarity metric (e.g., MSE) in the deterministic setting, or a distributional divergence (JSD, KL, MMD) for stochastic models;  $P_I$  is a distribution over interventions on macro-variables extended by  $\Phi$ ; and for a macro-intervention  $\nu = \text{do}(X = x)$ ,  $P_{\nu}$  denotes the distribution over compatible micro-interventions  $\mu = \text{do}(a^{-1}(X) = \mathbf{z})$  with  $\mathbf{z} \sim \tau_X^{-1}(x)$ . In  $\text{CAE}_{\downarrow}$  (top-down), macro-interventions are grounded to compatible micro-interventions via  $\tau_X^{-1}$ ; in  $\text{CAE}_{\uparrow}$  (bottom-up), micro-interventions are drawn directly and lifted to the macro level via  $\tau_X$ . Both arguments of  $D$  lie in the macro-variable range  $\mathcal{R}_Y$ , avoiding ill-posedness from  $\tau_Y$  being non-injective. Faithfulness is integrated in both variants:  $P_I$  includes interventions on  $\Phi$ -variables, which the proposed explanation claims are causally inert. Finally, we also define non-faithful variants:  $\text{CAE}_{\uparrow\text{NF}}$  and  $\text{CAE}_{\downarrow\text{NF}}$ .

**Relation to Zennaro et al. [2023b].** That work defines four related metrics; ISIL and IC are the closest to  $\text{CAE}_{\downarrow}$  and  $\text{CAE}_{\uparrow}$  respectively. Beyond the faithfulness consideration, three differences are worth noting. First, their metrics aggregate by supremum; ours use expectation, which is more tractable to estimate by sampling. Second, their framework is restricted to finite variable domains, whereas CAE handles continuous value maps directly. Third, their metrics intervene on a single variable at a time; Zennaro et al. [2023b] explicitly notes that this prevents compositionality of valid abstractions from being preserved. CAE supports joint multi-node interventions, restoring this property and probing interaction terms that single-variable interventions miss.

#### 3.1 Practical Estimation

Both variants of CAE use expectations amenable to Monte Carlo estimation. In our setting,  $\mathcal{E}$  is always deterministic, so there is no  $\mathcal{U}_{\mathcal{E}}$  to sample; the role of the exogenous noise  $u \in \mathcal{U}_{\mathcal{M}}$  is played by the intervention values drawn from  $P_I$ , which are passed identically to both models in each Monte Carlo sample. We discuss further key design choices below.

**Sampling macro-interventions ( $\text{CAE}_{\downarrow}$ ).** We draw  $N$  interventions  $\{\nu^{(i)}\}_{i=1}^N \sim P_I$ . To probe interaction terms, we support *multi-node interventions* by sampling a random subset of variables  $\mathcal{S} \subseteq \mathcal{V}^M$  of size  $|\mathcal{S}| \sim \text{Uniform}\{1, \dots, |\mathcal{V}^M|\}$  and jointly drawing intervention values  $\mathbf{x}_{\mathcal{S}}$ , where each component  $x_V$  for  $V \in \mathcal{S}$  is drawn independently and uniformly from the domain  $\mathcal{R}_V$  of the corresponding macro-variable. When a geometric domain is specified for a variable, intervention values are drawn by sampling directly from that subspace.

**Sampling compatible micro-interventions ( $\text{CAE}_{\downarrow}$ ).** For each macro-intervention  $\nu$  on variables  $\mathcal{S}$ , we sample a compatible micro-intervention by applying the inverse value map to each intervened variable: for  $V \in \mathcal{S}$  with intervened value  $x_V$ , we compute the preimage  $\tau_V^{-1}(x_V)$  and sample a micro-state from it. In discrete settings, this corresponds to the Moore-Penrose pseudo-inverse, giving uniform probability mass to all micro-states mapping to  $x_V$  [Zennaro et al., 2023a]. For continuous



systems, where the preimage is a subspace of the micro-state space, we sample uniformly from that subspace.

**Sampling micro-interventions** ( $\text{CAE}_{\dagger}$ ). We draw  $N$  micro-interventions  $\{\mu^{(i)}\}_{i=1}^N \sim P_I$  directly, using the same multi-node scheme as above but operating over micro-variable domains. The corresponding macro-intervention is obtained by applying  $\tau_X$  to the intervened micro-state.

**Testing faithfulness.** We test faithfulness by augmenting the intervention pool with a dummy  $\Phi$ -sentinel alongside the real macro-variables. At each sample,  $\Phi$  is included in the selected intervention targets with a configurable probability. When selected, independent noise is injected into every  $\phi \in \Phi$  simultaneously before executing  $\mathcal{M}$ : Gaussian noise with standard deviation  $\sigma_\phi$  for continuous variables, uniform  $\{-1, 0, 1\}$  perturbations for integer variables, and random bit-flips for Boolean variables. If the abstracted output shifts relative to the unperturbed run, the consistency error increases proportionally. The parameter  $\sigma_\phi$  controls the sensitivity of the test.

## 4 A Benchmark of Simulated Complex Systems

A central obstacle in evaluating explanation-validity metrics is the lack of naturally occurring systems paired with objective ground-truth explanations. Human judgment is an inadequate substitute, since users often prefer convincing explanations even when they are incomplete or misleading [Ancona et al., 2019, Herman, 2017, Petsiuk et al., 2018]. Existing benchmarks therefore typically evaluate desirable properties of explanation methods, such as sparsity, stability, localization, or causal robustness, rather than whether an explanation recovers a known valid high-level causal account [Karvonen et al., 2025, Poché et al., 2025, Liu et al., 2021, Huang et al., 2024].

We address this by constructing a benchmark of idealized complex systems whose high-level explanations are known and analytically justified at the relevant abstraction level. This enables two basic tests: a metric should assign near-zero error to correct explanations, and should detect controlled failures produced by invalid explanations. A key contribution of this work is the manual alignment of these heterogeneous systems with their high-level explanations integrated within a common framework. The benchmark spans many systems chosen to cover different abstraction types, yet all systems admit clean causal-abstraction representations and, more strongly, *constructive* causal abstractions. This provides a reusable resource and a nontrivial empirical demonstration of the breadth of the causal abstraction program. For each system, we provide the high-level model, low-level implementation, abstraction map, and manually crafted invalid contrastive scenarios. Full details are given in Appendix D; Table 1 summarizes the systems.

Table 1: **Benchmark systems.** Each row is a manually aligned ground-truth abstraction pair  $(\mathcal{M}, \mathcal{E}, \tau)$ . The benchmark spans discrete (D) and continuous (C) state spaces, static and dynamical regimes, and local, spatial, and global constructive maps. Rows shaded in light blue are dynamical systems. For every pair, we also provide invalid contrastive conditions by perturbing the model, the map, or the validity regime; full details are in Appendix D.

| System                  | Low level $\mathcal{M}$     | High level $\mathcal{E}$     | Constructive abstraction map $\tau$ |                      |
|-------------------------|-----------------------------|------------------------------|-------------------------------------|----------------------|
|                         |                             |                              | Type                                | Structure            |
| Gas                     | Lennard-Jones particles     | Ideal gas / VdW law          | C $\rightarrow$ C                   | global / aggregation |
| Predator-prey           | Agent-based model           | Lotka-Volterra ODE           | D $\rightarrow$ C                   | global / aggregation |
| Heat equation 1D        | Brownian particles          | 1D heat equation             | C $\rightarrow$ C                   | spatial / binning    |
| Heat equation 2D        | Phonon/lattice simulation   | 2D heat equation             | C $\rightarrow$ C                   | spatial / smoothing  |
| Ising model             | MD/Monte-Carlo simulator    | Nearest-neighbor Ising model | C $\rightarrow$ C                   | global / observables |
| Tracr transformer       | Compiled transformer states | Symbolic sort-rank program   | D $\rightarrow$ D                   | local / decoding     |
| Gene regulatory network | Multi-valued Boolean GRN    | Binary Wg-Fz rule            | D $\rightarrow$ D                   | local / threshold    |
| Logic circuit           | Wire-level Boolean circuit  | Multi-bit gate circuit       | D $\rightarrow$ D                   | local / functional   |
| Transistor circuit      | SPICE/MOSFET voltages       | Boolean gate network         | C $\rightarrow$ D                   | local / threshold    |
| MOS 6502 CPU            | Transistor/segment state    | Logic-gate state             | D $\rightarrow$ D                   | local / functional   |
| MOS 6502 CPU            | Logic-gate state            | ISA registers and flags      | D $\rightarrow$ D                   | local / functional   |
| MOS 6502 CPU            | Transistor/segment state    | ISA registers and flags      | D $\rightarrow$ D                   | local / functional   |

## 5 Results

We evaluate metrics along three dimensions in sequence: classification performance under high sampling, discrimination of structurally invalid abstractions in controlled experiments, and sampling efficiency as a function of model size.

**Benchmarking Metrics.** We begin by asking which metrics can, in principle, separate valid from invalid abstractions when sampling is not a constraint. We evaluate all metrics the asymptotic regime on every (valid, invalid) abstraction pair in the benchmark, with full results reported in Appendices A.1 and A.2. These results motivate two principled thresholds: after normalization, a metric should score at most 0.10 on a valid abstraction and at least 0.15 on an invalid one. Figure 1 uses these thresholds to assess the classification performance of each metric across the benchmark.

|                                             |           | OBSERVATIONAL |       |       |       |       |       |       |       |       |            | TEMPORAL |        |          |          |       | FUNCTIONAL |            |           |           |         | INFO. & REPR. |       |       | CAUSAL & INTERV. |         |         |            | CAUSAL ABSTRACTION |       |       |        |        |        |       |       |
|---------------------------------------------|-----------|---------------|-------|-------|-------|-------|-------|-------|-------|-------|------------|----------|--------|----------|----------|-------|------------|------------|-----------|-----------|---------|---------------|-------|-------|------------------|---------|---------|------------|--------------------|-------|-------|--------|--------|--------|-------|-------|
|                                             |           | MSE           | RMSE  | R     | PSD   | KL    | L1    | RMSE  | MMD   | JSIC  | VerDiscomp | TrajMSE  | DTW    | Autocorr | Spectral | SINDy | Sobol      | Infidelity | Rel. fid. | Mail. Co. | Struct. | Causal sens.  | IB    | CIB   | Complexity       | Probing | Symbion | Macro Inv. | BCC                | DCC   | IIA   | CAE 1w | CAE 1w | CAE 1w |       |       |
| Valid detection<br>score < 0.10 → valid     | Precision | 0.689         | 0.763 | 0.738 | 0.686 | 0.655 | 0.676 | 0.724 | 0.625 | 0.605 | 0.763      | 0.749    | 0.714  | 0.833    | 0.867    | 0.857 | 0.838      | 0.588      | 0.552     | 0.338     | 0.417   | 0.381         | -     | 0.500 | 0.492            | 0.522   | 0.723   | 0.566      | 0.538              | 0.500 | 0.642 | 0.775  | 0.791  | 0.789  | 0.805 |       |
|                                             | Recall    | 0.886         | 0.829 | 0.856 | 0.686 | 0.871 | 0.883 | 1.000 | 1.000 | 0.742 | 0.829      | 1.000    | 1.000  | 1.000    | 1.000    | 1.000 | 0.200      | 0.606      | 0.457     | 0.875     | 0.625   | 0.500         | 0.901 | 0.000 | 0.935            | 0.387   | 0.527   | 0.857      | 1.000              | 0.583 | 0.971 | 0.886  | 0.971  | 0.857  | 0.943 |       |
|                                             | F1        | 0.775         | 0.795 | 0.805 | 0.686 | 0.750 | 0.769 | 0.840 | 0.769 | 0.667 | 0.795      | 0.870    | 0.833  | 0.909    | 0.800    | 0.923 | 0.292      | 0.597      | 0.500     | 0.667     | 0.500   | 0.432         | -     | 0.344 | 0.644            | 0.444   | 0.615   | 0.682      | 0.700              | 0.538 | 0.773 | 0.827  | 0.872  | 0.822  | 0.868 |       |
| Invalid detection<br>score > 0.15 → invalid | Precision | 1.000         | 0.862 | 0.833 | 0.720 | 0.882 | 0.824 | 1.000 | 1.000 | 0.750 | 0.844      | 1.000    | 1.000  | 1.000    | 1.000    | 1.000 | 0.509      | 0.594      | 0.525     | 0.667     | 0.181   | 0.273         | 0.300 | 0.500 | 0.500            | 1.000   | 0.536   | 1.000      | 0.708              | 1.000 | 0.444 | 0.933  | 0.940  | 0.958  | 0.962 | 0.960 |
|                                             | Recall    | 0.514         | 0.714 | 0.571 | 0.534 | 0.484 | 0.500 | 0.616 | 0.371 | 0.387 | 0.426      | 0.700    | 0.600  | 0.700    | 0.300    | 0.031 | 0.020      | 0.074      | 0.411     | 0.258     | 0.002   | 0.180         | 1.000 | 0.914 | 0.022            | 0.536   | 0.500   | 0.000      | 0.071              | 0.333 | 0.400 | 0.406  | 0.657  | 0.714  | 0.486 |       |
|                                             | F1        | 0.678         | 0.781 | 0.678 | 0.600 | 0.625 | 0.622 | 0.765 | 0.542 | 0.511 | 0.721      | 0.824    | 0.750  | 0.824    | 0.442    | 0.069 | 0.030      | 0.055      | 0.264     | 0.367     | 0.222   | 0.247         | 0.644 | 0.062 | 0.516            | 0.647   | 0.311   | 0.133      | 0.381              | 0.560 | 0.800 | 0.780  | 0.820  | 0.800  |       |       |
| Applicability                               |           | 35/35         | 35/35 | 35/35 | 35/35 | 35/35 | 35/35 | 35/35 | 35/35 | 35/35 | 100/35     | 100/35   | 100/35 | 100/35   | 100/35   | 60/35 | 35/35      | 35/35      | 14/35     | 10/35     | 16/35   | 10/35         | 35/35 | 35/35 | 35/35            | 35/35   | 35/35   | 35/35      | 35/35              | 35/35 | 35/35 | 35/35  | 35/35  | 35/35  | 35/35 |       |

Figure 1: Precision, recall and F1-score of each metric, computed over all valid and invalid abstractions of the benchmark (higher is better). Applicability denotes the fraction of abstractions for which each metric was able to be evaluated; by design, many metrics are only applicable to a restricted class of systems. Metric normalization, implementation, and hyperparameters are reported in Appendix E.

Coverage (applicability) immediately eliminates a large subset of candidates. Temporal metrics (TrajMSE, DTW, Autocorr, Spectral, SINDy) are defined only for dynamical systems; Symbion requires finite discrete label spaces;  $C_p$  requires a linear proxy model with a well-defined noise floor; and structural deviation and the causal sensitivity index fail to produce scores on most benchmark systems. Among metrics with broad coverage, information-theoretic scores (IB and CIB Lagrangian) and several functional metrics, such as Sobol indices, structural deviation, and causal sensitivity indices, achieve mediocre to poor F1 on the valid task: their scores are sensitive to distributional details unrelated to causal consistency, so they assign non-trivial error even to valid abstractions. On the invalid task, the picture is mixed: some metrics attain reasonable precision, but at the cost of low recall. This combination of poor calibration and limited applicability immediately eliminates many candidate metrics from further consideration. Conversely, causal abstraction metrics consistently achieve among the highest F1 on both tasks and maximum coverage.

**Detection of Controlled Failure Cases.** Classification performance under unlimited sampling does not capture whether a metric can detect the specific structural failures that motivate causal abstraction. We therefore turn to six controlled experiments, each designed to expose a subtle violation that is observationally indistinguishable from the valid condition or manifests only under targeted interventions. The six settings are illustrated in Figure 2 and described in more detail in Appendix C. Figure 3 then reports, for each experiment and metric, whether a two-sample  $t$ -test over 100 runs reveals a statistically significant difference between the valid and invalid condition.

The results show that among the metrics that do apply, none of the observational, functional, or information-theoretic criteria achieve consistent discrimination across the six experiments. This is expected: each invalid abstraction is observationally indistinguishable from the valid one or fails only under specific interventional conditions, which are precisely the failure modes that causal abstraction was designed to detect. Within the causal abstraction family, results are more nuanced. IIA only succeeds in detecting the violations in Experiments 4 and 5. Both  $CAE_{NF}$  variants fail in Experiment 1 and 2; only the  $CAE$  variants successfully detect the violations.

Together, the benchmark results and controlled experiments narrow the candidate set to four metrics. IIA is excluded because, despite broad coverage, it fails to detect the violations in Experiments 3 and

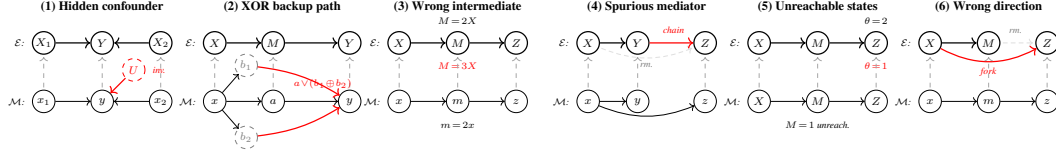


Figure 2: Causal graphs and abstraction maps for the six controlled experiments. Each panel shows  $\mathcal{E}$  (top) and  $\mathcal{M}$  (bottom). Gray dashed arrows are the abstraction map  $\tau$ . Red marks what changes in the *invalid* condition; light gray dashed arrows mark edges present in the valid condition that are removed in the invalid condition. Dashed gray circles are  $\phi$  variables (unmapped by  $\tau$ ); a dashed red circle is a  $\phi$  variable introduced only in the invalid condition.

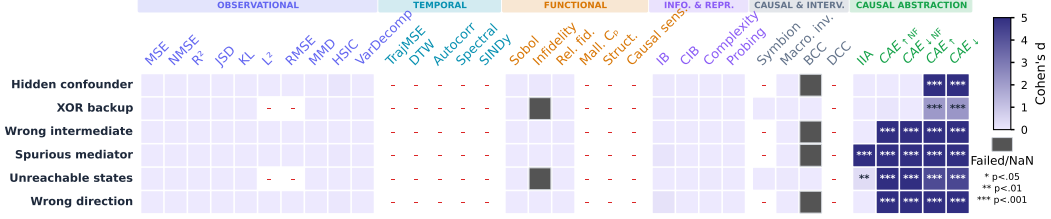


Figure 3: **Controlled failure modes.** Each cell reports whether a metric significantly separates the valid and invalid abstraction over 100 runs using a two-sample  $t$ -test. CAE is the only family that detects all six targeted invalidities.

310 6 where other causal abstraction metrics succeed. The remaining candidates are  $\text{CAE}_{\downarrow}$  and  $\text{CAE}_{\uparrow}$ , which  
311 we carry forward as the primary metrics. Their faithfulness-free counterparts,  $\text{CAE}_{\downarrow\text{NF}}$  and  $\text{CAE}_{\uparrow\text{NF}}$ , are  
312 retained as ablations to isolate the contribution of faithfulness testing.

313 **Statistical Power.** Having narrowed the candidate set to four metrics, we ask how many inter-  
314 ventions each requires to reliably detect an invalid abstraction. Convergence results, reported in  
315 Appendix 9, show that all four metrics stabilize within 30 interventions on nearly all benchmark  
316 systems. Figure 4 reports statistical power across selected systems and controlled experiments.

317 Statistical power converges to nearly 100% within 30 interventions for most systems and all four  
318 metrics. Two exceptions stand out. For the gas simulation, power increases more slowly, consis-  
319 tent with the higher variance noted above. Experiment 2 (XOR leak) illustrates the usefulness of  
320 faithfulness testing: metrics that include it detect the invalid abstraction as the  $n$  grows, while their  
321 faithfulness-free counterparts consistently fail. The noise condition on the GRN system illustrates  
322 the value of top-down sampling:  $\text{CAE}_{\downarrow}$  and  $\text{CAE}_{\downarrow\text{NF}}$  can reliably detect the invalid abstraction after  
323 25–30 interventions, while bottom-up metrics converge more slowly. For Experiment 5 (unreachable  
324 intermediate states), the picture reverses: faithfulness-aware metrics converge more slowly and exhibit  
325 lower power than plain  $\text{CAE}_{\text{NF}}$ .  $\text{CAE}_{\downarrow\text{NF}}$  already probes unreachable states by sampling from the full  
326 abstract label domain; faithfulness augmentation additionally injects perturbations into unmapped  $\Phi$

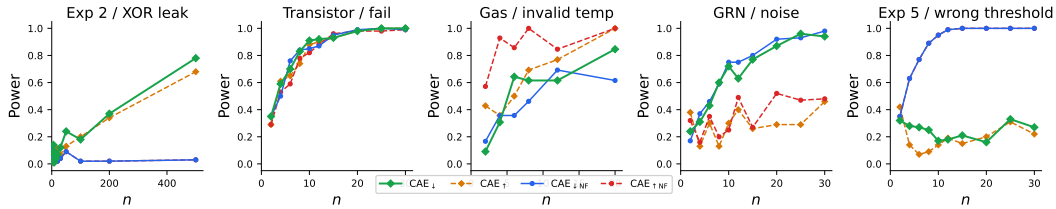


Figure 4: How quickly do metrics detect bad abstractions over good ones? For different numbers of sampled interventions  $n$ , we compute each metric 100 times on selected valid and invalid abstractions. We apply a two-sample T-test to the two sets of scores (valid and invalid) and report the fraction of runs for which the results differ significantly ( $p < 0.05$ ). Full results are given in appendix A.



variables, which occasionally push  $\mathcal{M}$  into the same micro-region as the unreachable states, inflating the valid-condition score and shrinking the discrimination gap. Faithfulness testing therefore does not miss the invalid abstraction here, but reduces specificity on the valid side.

**Scaling.** Power curves on fixed systems do not reveal how detection cost scales as the high-level model  $\mathcal{E}$ . To answer this question, we construct a controlled scaling experiment using Tracr [Lindner et al., 2023], a compiler that translates RASP programs into exact transformer weights. We compile a sort-rank program for each sequence length  $\ell \in \{2, 3, 4, 5, 6, 8, 10, 12, 15\}$ , yielding transformers of increasing size that compute the rank of each token in the input sequence. The valid high-level model ( $\mathcal{E}$ ) encodes the correct formula  $\text{rank}_i = |\{j \neq i : \text{token}_j < \text{token}_i\}|$  for all positions  $i$ , while the wrong model encodes a specific incorrect hypothesis: the minimum-valued token at position 0 always receives the maximum rank rather than rank 0. This error triggers exactly when  $\text{token}_0$  is the minimum element of the sequence, an event with probability  $1/\ell$  under uniform token sampling. The expected number of interventions to observe at least one triggering event therefore grows linearly with  $\ell$ , and analytic calculation gives  $n_{95\%} \approx 3\ell$  for 95% power.

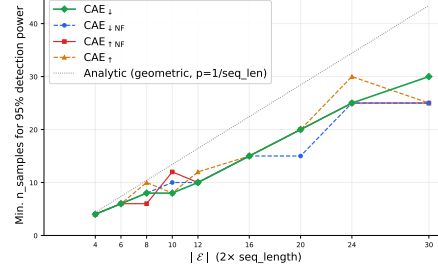


Figure 5: Minimum number of interventions required to reach 95% detection power as a function of  $|\mathcal{E}|$ , measured on Tracr sort-rank programs of increasing sequence length (100 runs per length). More details are provided in Appendix A.4.

Figure 5 shows the empirically measured required  $n$  for CAE and  $\text{CAE}_{\text{NF}}$  as a function of  $|\mathcal{E}| = 2\ell$  (token inputs plus rank outputs). All four metrics follow the predicted linear trend closely. This confirms that the sampling budget is governed primarily by the rarity of the triggering condition in the abstract domain, not by the internal complexity of the transformer. In practice, a wrong explanation that diverges from the correct one on a fraction  $p$  of inputs requires  $O(1/p)$  interventions to detect with fixed confidence: rare but systematic errors in proposed explanations remain detectable at a cost proportional to their rarity, independent of model depth or width.

## 6 Discussion

This paper asks when a high-level explanation of a low-level system should be trusted. Our benchmark turns this into an empirical stress test: given valid and invalid abstraction pairs, which metrics separate them? The main result is that observational agreement, distribution matching, representational similarity, compression, symbolic fit, and input-output sensitivity can be useful diagnostics, but not validity criteria. Across heterogeneous systems, causal-abstraction metrics are the most reliable. However, our results also show that operationalization matters: not all causal metrics succeed and different implementations fail in different ways. The proposed CAE metrics, which combine multi-node interventions with built-in faithfulness tests, are the only metrics that pass all tests. This makes CAE a natural objective for explanation discovery. Symbolic, gradient-based, evolutionary, and human-guided search procedures all need a criterion that rewards causally valid abstractions. In particular, future work should study adaptive intervention sampling to quickly detect invalid abstraction, CAE-based abstraction discovery, and community extensions of the benchmark. We release the benchmark and all metric implementations to support this direction.

### 6.1 Limitations

First, sample-level CAE assumes shared exogenous noise across abstraction levels. Our benchmark avoids this for deterministic macro-models, but stochastic micro-models are evaluated with independent stochastic realizations. Coupled stochastic abstractions remain future work. Second, the benchmark necessarily involves design choices that can be debated: systems, valid abstractions, invalid contrasts, and sampled interventions. We mitigate this by trying to span diverse domains and abstraction types, but future versions should include community-contributed systems and possibly blind evaluations. Finally, our power-scaling analysis is limited to a specific Tracr setting focused on explanation size. Yet, intervention requirements to distinguish an invalid explanation will depend on properties of the system, abstraction map, explanation size, and invalid contrast.

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## A Additional results

This section contains multiple additional figures for results described in the main body of the work, as well as a new experiment on convergence rates.

### A.1 Validity on Correct Abstractions

Figure 6 shows the scores produced by each baseline metric when applied to the correct abstractions in our benchmark.

|             |               | OBSERVATIONAL |       |       |       |       |                |       |       |       |            | TEMPORAL |       |          |          | FUNCTIONAL |        |            |           | INFO. & REPR. |         |             |       | CAUSAL & INTERV. |            |          | CAUSAL ABSTRACTION |       |       |                    |                    |                    |                    |       |       |
|-------------|---------------|---------------|-------|-------|-------|-------|----------------|-------|-------|-------|------------|----------|-------|----------|----------|------------|--------|------------|-----------|---------------|---------|-------------|-------|------------------|------------|----------|--------------------|-------|-------|--------------------|--------------------|--------------------|--------------------|-------|-------|
|             |               | MSE           | MMSE  | RE    | FSD   | KL    | L <sup>2</sup> | RMSE  | MMD   | JSIC  | VerDiscomp | TrajMSE  | DTW   | Autocorr | Spectral | zInfo      | Global | Infidelity | Rel. fid. | Mail. Co.     | Struct. | Causal sibs | IG    | CIB              | Complexity | Symbolic | Micro. int.        | DCC   | IIA   | CAE <sup>↑</sup> w | CAE <sup>↓</sup> w | CAE <sup>↑</sup> f | CAE <sup>↓</sup> f |       |       |
| Gas law     | Logic circuit | 0.000         | 0.000 | 0.000 | 0.000 | 0.000 | N/A            | N/A   | 0.000 | 0.000 | 0.000      | N/A      | N/A   | N/A      | N/A      | N/A        | 0.000  | 0.000      | 0.000     | N/A           | N/A     | N/A         | 0.678 | 0.000            | 0.000      | 0.000    | 0.000              | 0.000 | 0.000 | 0.000              | 0.000              | 0.000              | 0.000              | 0.000 | 0.000 |
|             | Transistor    | 0.000         | 0.000 | 0.000 | 0.000 | 0.000 | N/A            | N/A   | 0.000 | 0.000 | 0.000      | N/A      | N/A   | N/A      | N/A      | N/A        | 0.000  | 0.000      | 0.000     | N/A           | N/A     | N/A         | 0.618 | 0.000            | 0.000      | 0.000    | 0.000              | 0.000 | 0.000 | 0.000              | 0.000              | 0.000              | 0.000              | 0.000 | 0.000 |
|             | Ideal         | 0.000         | 0.000 | 0.000 | 0.000 | 0.000 | 0.000          | 0.000 | 0.000 | 0.000 | 0.000      | N/A      | N/A   | N/A      | N/A      | N/A        | 0.000  | 0.000      | 0.000     | N/A           | N/A     | N/A         | 0.678 | 0.000            | 0.000      | 0.000    | 0.000              | 0.000 | 0.000 | 0.000              | 0.000              | 0.000              | 0.000              | 0.000 | 0.000 |
|             | VdW           | 0.000         | 0.000 | 0.000 | 0.000 | 0.000 | 0.000          | 0.000 | 0.000 | 0.000 | 0.000      | N/A      | N/A   | N/A      | N/A      | N/A        | 0.000  | 0.000      | 0.000     | N/A           | N/A     | N/A         | 0.678 | 0.000            | 0.000      | 0.000    | 0.000              | 0.000 | 0.000 | 0.000              | 0.000              | 0.000              | 0.000              | 0.000 | 0.000 |
|             | Predator-prey | 0.000         | 0.000 | 0.000 | 0.000 | 0.000 | 0.000          | 0.000 | 0.000 | 0.000 | 0.000      | 0.000    | 0.000 | 0.000    | 0.000    | 0.000      | 0.000  | 0.000      | 0.000     | 0.000         | 0.000   | 0.000       | 0.000 | 0.000            | 0.000      | 0.000    | 0.000              | 0.000 | 0.000 | 0.000              | 0.000              | 0.000              | 0.000              | 0.000 |       |
| Heat eq. 1D | Heat eq. 2D   | 0.000         | 0.000 | 0.000 | 0.000 | 0.000 | 0.000          | 0.000 | 0.000 | 0.000 | 0.000      | 0.000    | 0.000 | 0.000    | 0.000    | 0.000      | 0.000  | 0.000      | 0.000     | 0.000         | 0.000   | 0.000       | 0.000 | 0.000            | 0.000      | 0.000    | 0.000              | 0.000 | 0.000 | 0.000              | 0.000              | 0.000              | 0.000              | 0.000 |       |
|             | Ising model   | 0.000         | 0.000 | 0.000 | 0.000 | 0.000 | 0.000          | 0.000 | 0.000 | 0.000 | 0.000      | 0.000    | 0.000 | 0.000    | 0.000    | 0.000      | 0.000  | 0.000      | 0.000     | 0.000         | 0.000   | 0.000       | 0.000 | 0.000            | 0.000      | 0.000    | 0.000              | 0.000 | 0.000 | 0.000              | 0.000              | 0.000              | 0.000              | 0.000 |       |
| Tracr       | GRN           | 0.000         | 0.000 | 0.000 | 0.000 | 0.000 | 0.000          | 0.000 | 0.000 | 0.000 | 0.000      | 0.000    | 0.000 | 0.000    | 0.000    | 0.000      | 0.000  | 0.000      | 0.000     | 0.000         | 0.000   | 0.000       | 0.000 | 0.000            | 0.000      | 0.000    | 0.000              | 0.000 | 0.000 | 0.000              | 0.000              | 0.000              | 0.000              | 0.000 | 0.000 |
|             | gate+ISA      | 0.000         | 0.000 | 0.000 | 0.000 | 0.000 | 0.000          | 0.000 | 0.000 | 0.000 | 0.000      | 0.000    | 0.000 | 0.000    | 0.000    | 0.000      | 0.000  | 0.000      | 0.000     | 0.000         | 0.000   | 0.000       | 0.000 | 0.000            | 0.000      | 0.000    | 0.000              | 0.000 | 0.000 | 0.000              | 0.000              | 0.000              | 0.000              | 0.000 |       |
| CPU         | trans+gate    | 0.000         | 0.000 | 0.000 | 0.000 | 0.000 | 0.000          | 0.000 | 0.000 | 0.000 | 0.000      | 0.000    | 0.000 | 0.000    | 0.000    | 0.000      | 0.000  | 0.000      | 0.000     | 0.000         | 0.000   | 0.000       | 0.000 | 0.000            | 0.000      | 0.000    | 0.000              | 0.000 | 0.000 | 0.000              | 0.000              | 0.000              | 0.000              | 0.000 | 0.000 |

Figure 6: Error scores obtained by baseline metrics when applied to valid abstractions across benchmark systems. Scores are normalized to the  $[0, 1]$  range, where 0 denotes no error and 1 maximal error. Gray cells denote computation failure or NaN results for the metric.

### A.2 Discrimination Power on Invalid Abstractions

We report in Figure 7 the results of the discrimination measurement in 5, applied to all systems (including controlled experiments) and metrics.

### A.3 Power Curves

Figure 8 contains the full results of the power experiments in 5, computed across all invalid abstractions for all systems.

Figure 9 shows how the score estimates produced by each causal metric stabilize for different numbers of sampled interventions.

Estimates stabilize quickly across most systems: 30 sampled interventions are generally sufficient to obtain a stable score, with the gas simulation generally requiring more samples due to the higher variance of the molecular dynamics estimator. No metric stands out as substantially slower to converge than the others. However, differences between sampling directions emerge. For the GRN with a reversed causal rule, top-down metrics ( $CAE_{\downarrow}$  and  $CAE_{\downarrow NF}$ ), which intervene directly on the abstract states that expose the mis-specification, score higher than their bottom-up counterparts, which only draw micro-states reachable from the input distribution. For the GRN noise condition, bottom-up metrics score higher than top-down ones on valid abstractions: sampling generates interventions near the boundaries of the value-map subspaces, and noisy outputs of  $\mathcal{M}$  are more likely to be misclassified after abstraction. Top-down sampling stays in the interior of each subspace, reducing the noise. The separation is notably strong in Experiment 1 (hidden confounder):  $CAE_{NF}$  variants score near zero, while faithfulness-augmented variants detect the invalid abstraction (score near one). This demonstrates that faithfulness testing is critical, independently of the sampling direction.

### A.4 Tracr power curves

We provide here extended results for the Tracr sort-rank scaling experiment.



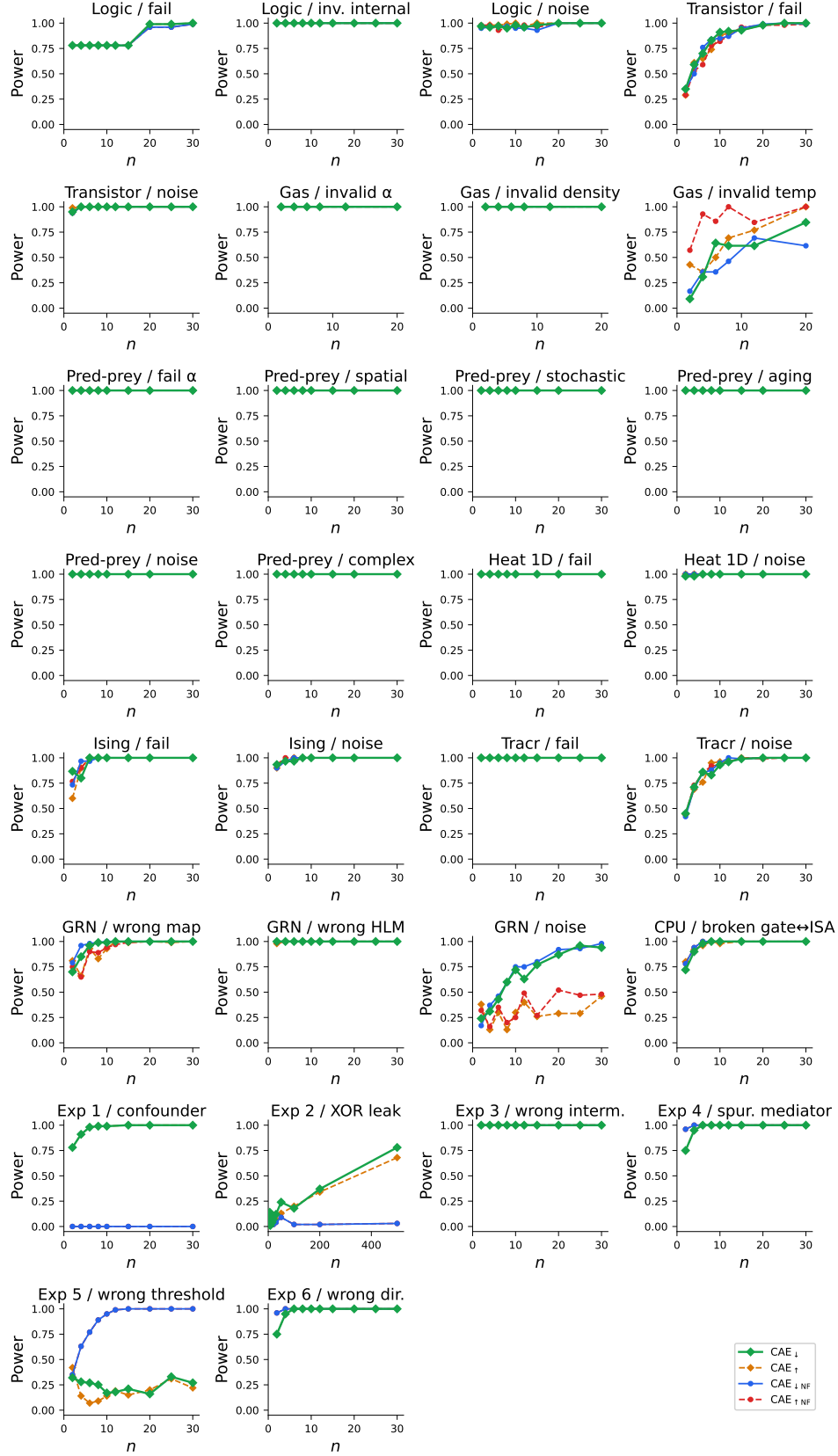


Figure 8: How quickly do metrics detect bad abstractions over good ones? For different numbers of sampled interventions  $n$ , we compute each metric 100 times on selected valid and invalid abstractions. We apply a two-sample T-test to the two sets of scores (valid and invalid) and report the fraction of runs for which the results differ significantly ( $p \leq 0.05$ ).

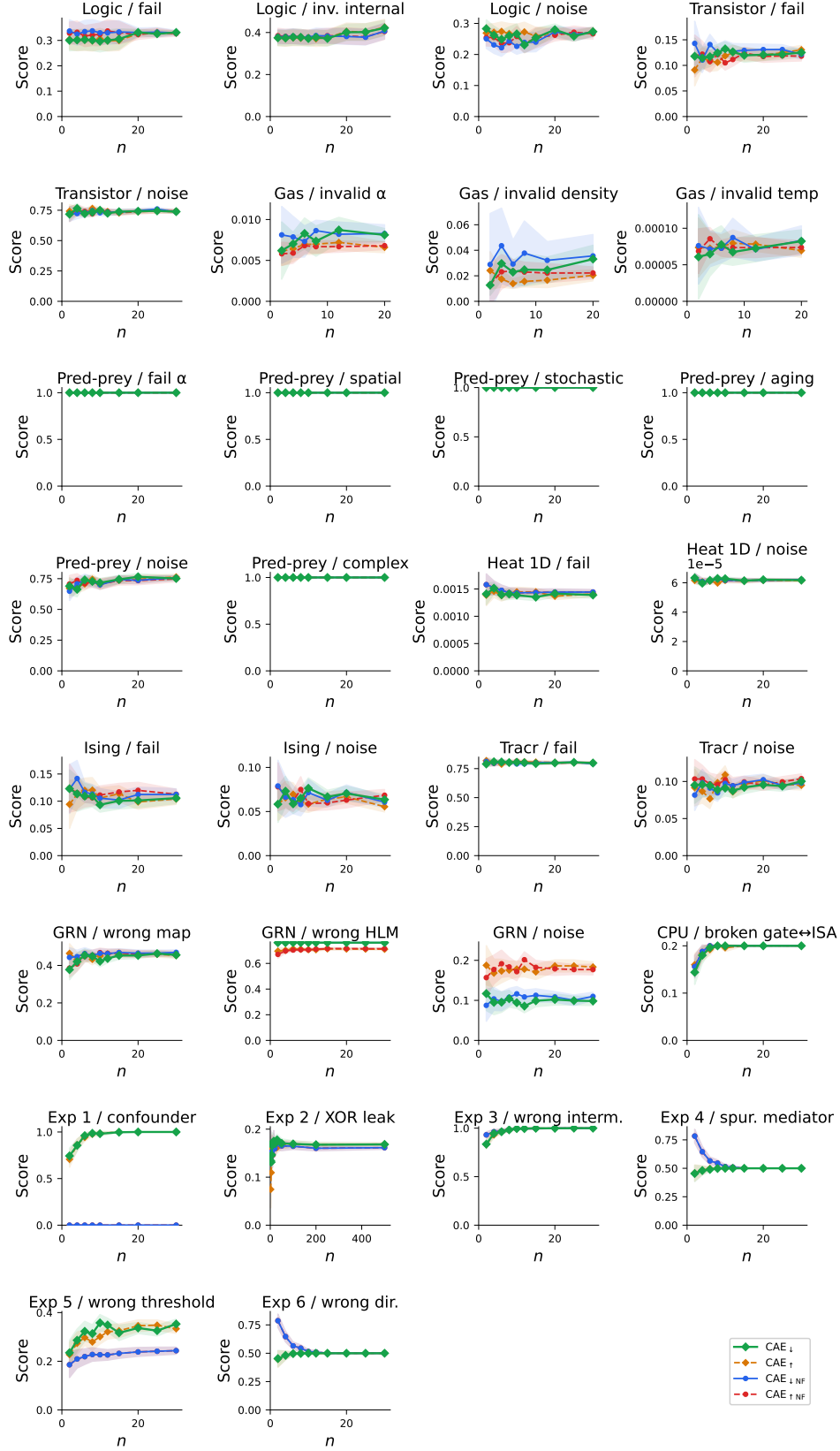


Figure 9: Convergence rate: For different numbers of sampled interventions  $n$ , we compute each metric 100 times on selected invalid abstractions.

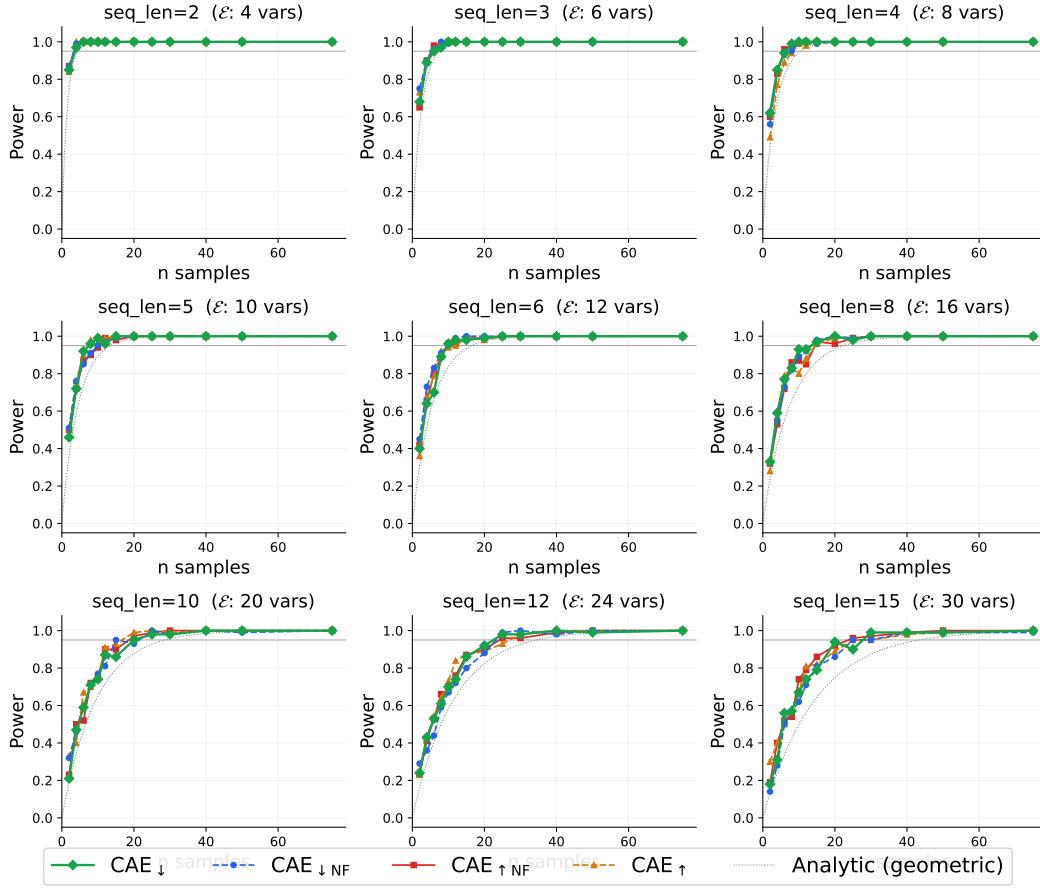


Figure 10: Detection power vs. the number of sampled interventions  $n$  across different sequence lengths for the Tracr sort-rank programs. The curves show the empirical probability of successfully detecting the invalid condition (where  $\text{rank}_0 = \ell - 1$  when  $\text{token}_0$  is the minimum).

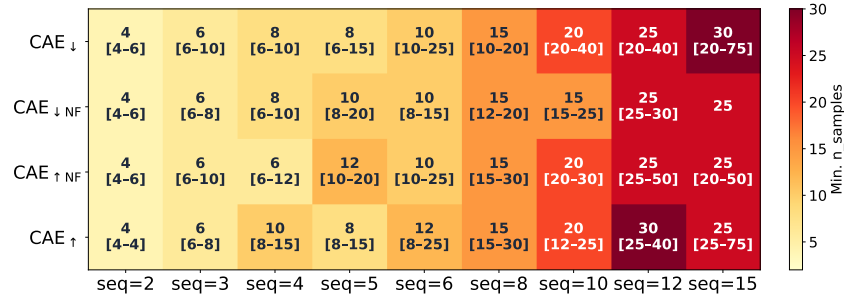


Figure 11: Minimum number of sampled interventions required to achieve 95% detection power on the Tracr sort-rank experiment, evaluated across different metrics (rows) and sequence lengths (columns). Numbers between brackets represent Wilson confidence intervals.



2019, Landajuela et al., 2022], where model selection is performed by minimizing reconstruction error on held-out data.

**Complexity-regularized accuracy.** Accuracy-only criteria permit arbitrarily complex explanations that overfit the observed data. Standard practice augments the reconstruction objective with a complexity penalty  $\Omega(\mathcal{E})$ , yielding criteria such as AIC [Akaike, 2003], BIC [Schwarz, 1978], and minimum description length [Mangan et al., 2017, Dong and Zhong, 2025]. *Mallow’s  $C_p$*  statistic [Vallejo et al., 2014, Beula et al., 2024, Lee and Ghosh, 2009] provides a related bias–variance decomposition, penalizing the number of free parameters:  $C_p = \text{RSS}/\hat{\sigma}^2 - n + 2p$ , where RSS is the residual sum of squares,  $p$  the number of parameters, and  $\hat{\sigma}^2$  an estimate of noise variance. In our setting,  $C_p$  is applied to a linear proxy fitted between the predictions of  $\mathcal{E}$  and the outputs of  $\mathcal{M}$  so that it simultaneously penalizes residual mismatch and model complexity in a cross-model comparison:  $p$  is set to twice the number of variables in  $\mathcal{E}$  (one slope and one intercept per variable), and  $\sigma^2$  is estimated from the residuals of this proxy. Physics-inspired approaches embed domain-specific inductive biases, such as dimensional homogeneity, separability, known invariances, directly into the admissible hypothesis class, as in AI Feynman [Udrescu and Tegmark, 2020] and related frameworks [Tenachi et al., 2023, Imai Aldeia et al., 2025]. We also employ a *SINDy-style evaluator* [Brunton et al., 2016], which scores an explanation by how well its governing equations predict the empirical trajectory one epoch ahead. High residual indicates that the governing equations of  $\mathcal{E}$  do not faithfully reproduce the system’s epoch-level dynamics.

**Distributional equivalence.** When  $\mathcal{M}$  is stochastic or the output of interest is a distributional quantity, pointwise comparison is insufficient. Observational validity is then defined by proximity between the output marginals  $P_{\mathcal{E}(u)}$  and  $P_{\mathcal{M}(u)}$ . The *KL divergence*  $D_{\text{KL}}(P_{\mathcal{M}}\|P_{\mathcal{E}})$  and the symmetric *Jensen–Shannon divergence*  $D_{\text{JS}}$  measure discrepancy in a model-based sense. The *maximum mean discrepancy*  $\text{MMD}^2(\mathcal{M}, \mathcal{E}) = \|\mu_{P_{\mathcal{M}}} - \mu_{P_{\mathcal{E}}}\|_{\mathcal{H}}^2$ , where  $\mu_P$  is the kernel mean embedding of  $P$  in a reproducing kernel Hilbert space  $\mathcal{H}$  [Gretton et al., 2012], provides a nonparametric alternative. The *Hilbert–Schmidt independence criterion* (HSIC) [Gretton et al., 2005] evaluates statistical dependence between residuals  $\mathcal{M}(u) - \mathcal{E}(u)$  and the input  $u$ ; a low HSIC indicates that errors are independent of the input, reflecting global rather than locally compensating agreement. These criteria subsume the *docking* paradigm in agent-based modeling [Axtell et al., 1996, Collins et al., 2024], where models are accepted as equivalent when they reproduce the same aggregate distributional patterns.

**Temporal fidelity.** For dynamical systems only, observational validity must be assessed over entire trajectories rather than marginal snapshots. *Trajectory MSE* measures the average squared deviation between state sequences  $\{x_t^{\mathcal{M}}\}$  and  $\{x_t^{\mathcal{E}}\}$  over a common time horizon. *Dynamic time warping* (DTW) [Sakoe and Chiba, 1978] generalizes this to temporally misaligned trajectories by computing the minimum-cost elastic alignment:  $\text{DTW}(\mathbf{x}^{\mathcal{M}}, \mathbf{x}^{\mathcal{E}}) = \min_{\pi} \sum_{(i,j) \in \pi} d(x_i^{\mathcal{M}}, x_j^{\mathcal{E}})$ , where  $\pi$  ranges over admissible warping paths. *Temporal autocorrelation matching* assesses whether  $\mathcal{E}$  reproduces the autocorrelation structure  $\rho^{\mathcal{M}}(\tau) = \mathbb{E}[x_t x_{t+\tau}]$  of  $\mathcal{M}$  across lags  $\tau$ ; failure indicates that  $\mathcal{E}$  does not capture the system’s memory structure. *Spectral analysis* compares the power spectral densities obtained via the Fourier transform of the respective autocorrelation functions [Percival and Walden, 1993]; agreement in dominant frequencies, harmonic structure, and spectral entropy indicates that the explanation captures the characteristic temporal modes of  $\mathcal{M}$ .

**Limitation: equifinality.** All observational criteria share a fundamental limitation: they are insensitive to the causal mechanisms generating the data. Distinct explanations may induce indistinguishable output distributions, a phenomenon termed *equifinality* [Valogianni and Padmanabhan, 2022, Collins et al., 2024]. Observationally valid models can violate the causal structure of  $\mathcal{M}$  while remaining empirically accurate, leading to explanations that are unstable under distribution shift [Ghorbani et al., 2019, Kindermans et al., 2019, Zhang et al., 2025] and fragile to mechanistic perturbations [Méloux et al., 2025b,a, Katende, 2025].

## 871 B.2 Functional Validity

*Functional validity* strengthens observational equivalence by requiring that  $\mathcal{E}$  reproduce not only the outputs of  $\mathcal{M}$  but also its *input–output response profile*: how outputs vary as inputs are systematically modified. These criteria are most naturally applied to static or quasi-static input–output systems, though several generalizations to dynamical settings exist.

**Variance decomposition and global sensitivity.** *Variance decomposition* decomposes the total output variance  $\text{Var}[\mathcal{M}(u)]$  into contributions attributable to individual inputs and their interactions via the ANOVA expansion [Efron and Stein, 1981, Saltelli et al., 1999]. *Sobol sensitivity indices* [Sobol', 2001] formalize this as  $S_i = \text{Var}[\mathbb{E}[\mathcal{M}(u) \mid u_i]] / \text{Var}[\mathcal{M}(u)]$  (first-order) and analogously for higher-order terms (interactions between input coordinates). Under this validity criterion,  $\mathcal{E}$  is deemed valid if its Sobol indices  $\hat{S}_i$  agree with those of  $\mathcal{M}$  up to an acceptable tolerance, ensuring global functional coherence in how uncertainty in each input propagates to the output. Related global sensitivity methods including FAST [Cukier et al., 1973, Saltelli et al., 1999] and Morris screening [Morris, 1991] provide computationally cheaper approximations to the same underlying sensitivity structure.

**Local attribution fidelity.** A complementary notion evaluates agreement between  $\mathcal{E}$  and  $\mathcal{M}$  in a neighborhood of a fixed input  $u_0$ . The *infidelity* metric Yeh et al. [2019] is defined as

$$\text{INFD}(\mathcal{E}, u_0) = \mathbb{E}_{\delta u} \left[ \left( \delta u^\top \phi(\mathcal{E}, u_0) - [\mathcal{M}(u_0) - \mathcal{M}(u_0 - \delta u)] \right)^2 \right],$$

where  $\phi(\mathcal{E}, u_0)$  is the attribution vector of  $\mathcal{E}$  at  $u_0$  and  $\delta u$  is drawn from a perturbation distribution. Low infidelity indicates that the explanation’s attributions accurately predict the actual output changes of  $\mathcal{M}$  under local perturbations. We adapt infidelity to compare output sensitivities between  $\mathcal{E}$  and  $\mathcal{M}$  directly, without requiring attribution vectors. LIME Ribeiro et al. [2016] constructs a locally faithful surrogate by minimizing a locally weighted reconstruction error around  $u_0$ , and SHAP Lundberg and Lee [2017] defines attributions via Shapley values, which satisfy a consistent axiomatic decomposition of the output into per-feature contributions; both can be interpreted as proxies for local attribution fidelity.

**Structural deviation and causal sensitivity index.** While standard sensitivity indices quantify statistical influence under passive sampling, a complementary family of metrics evaluates the sensitivity of explanatory validity to the structural parameters of  $\mathcal{E}$  itself [Katende, 2025]. The *structural deviation* metric perturbs each parameter of  $\mathcal{E}$  by a small amount and measures the resulting change in alignment score, identifying which parameters are most important for the explanation’s validity. The *causal sensitivity index* applies a harder test: each parameter is zeroed out in turn, and the drop in alignment score is recorded, providing a global decomposition of validity across the components of  $\mathcal{E}$ . Together, these metrics support model diagnosis by distinguishing robust explanatory components from those whose perturbation or removal would invalidate the abstraction.

**Counterfactual and relational fidelity.** *Counterfactual validity* [Wachter et al., 2017] requires that  $\mathcal{E}$  reproduce the minimal perturbations  $\delta u$  that change the class label or output regime of  $\mathcal{M}$ : the decision boundaries of  $\mathcal{E}$  must coincide with those of  $\mathcal{M}$  in the relevant neighborhood. *Relational fidelity* [Collins et al., 2024] measures the Pearson correlation between  $\Delta_{\mathcal{E}}(u, u') = \mathcal{E}(u') - \mathcal{E}(u)$  and  $\Delta_{\mathcal{M}}(u, u') = \alpha(\mathcal{M}(u')) - \alpha(\mathcal{M}(u))$  over sampled pairs  $(u, u')$ , computed independently per output dimension and averaged. A score of 0 (after normalization) indicates perfect agreement in relational structure.

**Limitation: structural underdetermination.** Functional criteria remain agnostic to internal mechanisms. Distinct models can exhibit identical global sensitivity profiles, local attributions, and counterfactual boundaries while relying on entirely different underlying structures. Functional validity therefore does not uniquely identify the causal organization of  $\mathcal{M}$ , and multiple mechanistically incompatible explanations may be simultaneously admissible.

### B.3 Information-Theoretic and Representational Validity

Another class of criteria requires that  $\mathcal{E}$  reproduce not only the input–output behavior of  $\mathcal{M}$  but also the structure of its internal representations: how information is encoded, compressed, and transmitted across the system.

**Representational alignment and probing accuracy.** A first approach tests whether the abstract variables posited by  $\mathcal{E}$  are linearly decodable from the internal states of  $\mathcal{M}$ . In the case of neural networks, given a representation  $h(u) \in \mathbb{R}^d$  extracted from a nominated layer of  $\mathcal{M}$ , *probing accuracy* measures the performance of a linear classifier or regressor trained to predict an explanatory variable from  $h(u)$  [Alain and Bengio, 2018]. For real neural networks, neuroscience commonly

refers to these practices as *multivariate pattern analysis* [Haxby et al., 2001, Norman et al., 2006] or *brain signatures* [Wager et al., 2013, Kragel et al., 2018]. High probing accuracy indicates that  $z$  is explicitly represented in  $\mathcal{M}$ ’s internal geometry [Ravichander et al., 2021, Belinkov, 2022]. For systems with continuous identity value maps, where all samples share a single abstract label, probing accuracy is trivially maximal and reflects the smoothness of the micro-model’s representation rather than the discriminability of abstract categories. Supervised concept-level probing is instantiated by Concept Activation Vectors [Kim et al., 2018]; unsupervised variants based on sparse autoencoders (SAEs) [Cunningham et al., 2023, Bricken et al., 2023] recover a dictionary of monosemantic features without requiring labeled concepts. Representational similarity analysis (RSA) [Kriegeskorte et al., 2008] generalizes this by comparing representational dissimilarity matrices between  $\mathcal{E}$  and  $\mathcal{M}$ , without committing to a specific linear structure [Schrimpf et al., 2021, Aw et al., 2024, Ryskina et al., 2025].

**Information bottleneck and causal information bottleneck Lagrangians.** The *information bottleneck* (IB) framework characterizes valid representations as those achieving an optimal trade-off between compression of the input and preservation of information relevant to the output [Tishby et al., 2000]. However, standard IB relies on observational mutual information, rendering it sensitive to spurious correlations in the input distribution. The *causal information bottleneck* (CIB) [Simoes et al., 2025] strengthens this by replacing observational relevance with causal relevance measured under interventions over inputs, so that  $\mathcal{E}$  is assessed by its capacity to preserve causally invariant predictive information. Operationally, the IB Lagrangian is evaluated using the observational mutual information  $I(\mathcal{E}(u); \mathcal{M}(u))$ , whereas the CIB Lagrangian replaces this with the interventional quantity  $I_c(Y \mid \text{do}(T))$ ; this distinction is relevant when the input distribution contains spurious correlations that increase the apparent observational relevance of  $\mathcal{E}$ .

**Complexity shift.** *Algorithmic information dynamics* (AID) [Zenil et al., 2019, 2020, 2023] evaluates explanations in terms of changes in algorithmic complexity induced by interventions. The *complexity shift* metric is defined as the discrepancy between the change in Kolmogorov complexity  $K$  of the output under perturbation of  $\mathcal{M}$  versus the corresponding change in  $\mathcal{E}$ :

$$\Delta K_{\mathcal{M}}(u, \delta u) = K(\mathcal{M}(u + \delta u)) - K(\mathcal{M}(u)), \quad \text{CS}(\mathcal{E}) = \mathbb{E}[\lvert \Delta K_{\mathcal{M}} - \Delta K_{\mathcal{E}} \rvert].$$

Low complexity shift indicates that  $\mathcal{E}$  produces mechanistically comparable transformations under perturbation. In practice, Kolmogorov complexity is approximated via compression algorithms or block decomposition methods [Zenil et al., 2019], introducing estimation variance.

**Limitation: mechanistic underdetermination.** Information-theoretic criteria capture *what information* is present and how it is organized across variables, but not *how* that information is causally transformed by the system’s dynamics. Distinct mechanisms can yield identical probing accuracies, identical IB optima, and identical complexity-shift profiles [Méloux et al., 2025a]. Moreover, most of these criteria, except in their explicitly causal variants, are grounded in observational regimes [Teney et al., 2022].

## B.4 Causal and Interventional Validity

The strictest class of criteria requires that  $\mathcal{E}$  correctly reproduce the behavior of  $\mathcal{M}$  under *active interventions*. Informally, let  $\mathcal{I}$  be a class of interventions on  $\mathcal{M}$ , and let  $\mathcal{M}^{(i)}$  and  $\mathcal{E}^{(i)}$  denote the respective post-intervention systems for  $i \in \mathcal{I}$ . Then  $\mathcal{E}$  is *interventionally valid* relative to  $\mathcal{I}$  if  $P_{\mathcal{M}^{(i)}}(Y) \approx P_{\mathcal{E}^{(i)}}(\hat{Y})$ ,  $\forall i \in \mathcal{I}$ , where  $Y$  and  $\hat{Y}$  are corresponding output variables under a fixed variable correspondence. Unlike observational criteria, this requirement is closed under the causal operator: agreement must hold across a family of intervention regimes.

**Behavioral causal consistency and macroscopic invariance.** A prerequisite for interventional validity is that the abstraction map  $\tau$  must be well-defined with respect to the dynamics: distinct micro-states sharing the same abstract label must produce the same abstract output under intervention, regardless of the micro-realization. In the Neural Causal Model framework [Grosse-Wentrup et al., 2024], this is formalized as *behavioral causal consistency* (BCC): for every pair  $x_1, x_2 \in \tau^{-1}(c)$ ,  $P(B[t] \mid \text{do}(X[t] = x_1)) = P(B[t] \mid \text{do}(X[t] = x_2))$ . The same requirement appears independently in statistical physics under the name *macroscopic invariance*: a coarse-graining

is admissible only when all micro-configurations within a macro-state are observationally equivalent at the macro-level [Fisher, 1998]. BCC and macroscopic invariance differ in scope: macroscopic invariance tests each variable in isolation, intervening on a single variable while leaving others unspecified, whereas BCC tests each variable within a complete system state drawn from the joint distribution. Operationally, these are assessed by sampling pairs of micro-states with the same abstract label and verifying that their abstracted outputs of  $\mathcal{M}$  agree under intervention.

**Dynamic causal consistency.** The *dynamic causal consistency* (DCC) criterion, introduced in the Neural Causal Model framework for dynamical systems [Grosse-Wentrup et al., 2024], requires that interventions on  $\mathcal{E}$  correctly predict the next-step internal states of  $\mathcal{M}$ . Formally, DCC is satisfied if  $\mathcal{E}^{(i)}(u_t) \approx h_{\mathcal{M}}(u_{t+1})$  for all interventions  $i \in \mathcal{I}$  and time steps  $t$ , where  $h_{\mathcal{M}}(\cdot)$  denotes the corresponding internal state of  $\mathcal{M}$ . DCC thus enforces step-wise causal coherence between the explanation’s dynamics and those of the target system. It is a dynamical extension of behavioral causal consistency (BCC): where BCC requires that same-label micro-states produce identical behavioral outputs under intervention, DCC additionally requires that micro-states sharing the same abstract label at time  $t$  must also do so at  $t + 1$  under the system dynamics. We evaluate this using exact label matching on the abstracted next state.

**Constructive and formal verification.** The strongest guarantees arise in settings where  $\mathcal{E}$  can be analytically derived from or formally verified against  $\mathcal{M}$ . In physics, renormalization group methods and effective field theory [Fisher, 1998, Georgi, 1993] construct coarse-grained explanations that provably preserve interventional predictions. In computer science, formal verification [Woodcock et al., 2009] establishes that an implementation  $\mathcal{M}$  satisfies a high-level specification  $\mathcal{E}$  for *all* inputs, yielding universal behavioral equivalence guarantees. *Symbion* [Gritti et al., 2020] operationalizes a weaker but practically tractable version of this: it enforces consistency between concrete executions of  $\mathcal{M}$  and symbolic executions of  $\mathcal{E}$  at each program step, grounding abstract explanatory claims in observed concrete behavior. The validity score assigned by Symbion corresponds to the fraction of execution steps at which the symbolic and concrete states agree. In our evaluation, we implement a discrete analog suited to finite abstract domains: we exhaustively enumerate all combinations of abstract input labels, execute both models, and measure the fraction of input combinations for which they disagree on any output. In causal agent-based modeling [Valogianni and Padmanabhan, 2022], explanations are constructed to satisfy domain-theoretic structural constraints.

**Mechanistic interpretability.** Developed primarily in AI, mechanistic interpretability (MI) aims to reverse-engineer networks into human-interpretable algorithms [Olah et al., 2020]. MI assumes that for a given behavior of interest, a sparse subset of the network executes the relevant algorithm. This subset, often referred to as a *circuit*, is a key research target of modern AI interpretability methods Vig et al. [2020], Meng et al. [2022], Monea et al. [2024], Kramár et al. [2024], Conmy et al. [2023], Geva et al. [2023], Syed et al. [2023]. It is loosely inspired by neuroscience, which also seeks to uncover neural circuits underlying observed behaviors Yuste [2008]. A circuit is valid if it reproduces the network output under interventions on the computation, whether they are part or not of the circuit [Hanna et al., 2024]. *Intervention interchange accuracy* (IIA) [Geiger et al., 2022b] generalizes this perspective: it requires that pairwise interchange interventions on the network’s internal representations and the high-level produce consistent outputs. Unlike IC and its variants, IIA evaluates interventional consistency without requiring a fully specified abstraction map over variables.

**Computational mechanics and  $\epsilon$ -machines.** A more general perspective is provided by computational mechanics, in which valid explanations correspond to *causally closed* macro-level representations. The  $\epsilon$ -machine framework [Crutchfield and Young, 1989, Shalizi and Crutchfield, 2001] defines the canonical valid explanation as the minimal sufficient statistic for predicting  $\mathcal{M}$ : the set of *causal states*  $\mathcal{S}$  such that future behavior is conditionally independent of the past given  $\mathcal{S}$ . An explanation is valid under this criterion if its state space is isomorphic to the  $\epsilon$ -machine of  $\mathcal{M}$ , ensuring that no micro-level information would improve predictive accuracy. Related perspectives interpret valid explanations as *programs implemented* by the underlying system, whose macro-level dynamics are self-contained and interventionally sufficient [Rosas et al., 2024].

**Limitations** While causal and interventional criteria provide the strongest notion of validity, existing formulations face practical limitations in complex, stochastic settings. Constructive approaches



require strong domain-specific assumptions. Mechanistic interpretability operates at the wrong level of abstraction, viewing low-level components as variables of the high-level explanation and, thus, suffer from causal overdetermination [McGrath et al., 2023, Méloux et al., 2025a]. Finally, computational mechanics and  $\epsilon$ -machines are subsumed by the more general framework of causal abstraction.

## C Controlled Experiments

We describe below the six controlled experiments from Figure 2. These highlight subtle ways in which abstraction can fail: aside from the strongest causal abstraction metrics, baseline metrics generally fail to detect the errors in these claimed abstractions.

**Experiment 1: Hidden confounder.** The macro-model posits that initial prey and predator populations determine the final populations (Lotka-Volterra dynamics).  $\mathcal{M}$  introduces an unmapped resource variable (a phi variable) that multiplicatively scales the final prey population output by a factor of  $\gamma \cdot r$ , where  $\gamma$  is the coupling strength and  $r$  is the resource level. This creates a hidden confounder that the macro-model cannot account for. A valid metric should detect that the abstraction fails under perturbations to this resource variable. In the valid condition, the resource coupling is set to zero ( $\gamma = 0$ ), making the resource variable inert.

**Experiment 2: XOR backup path.** The macro-model posits  $X \rightarrow M \rightarrow Y$ , with two phi variables  $b_1, b_2$  (declared causally inert). In the valid  $\mathcal{M}$ , the XOR path is structurally disconnected ( $y = a = x$ ), so perturbing  $b_1$  and  $b_2$  has no effect regardless of their values. In the invalid  $\mathcal{M}$ ,  $Y$  is computed as  $a \vee (b_1 \oplus b_2)$ , where  $a$  is the micro-variable corresponding to  $M$ : a hidden XOR backup path to  $Y$  fires only when the phi variables are independently perturbed. This experiment specifically tests faithfulness: structural metrics that do not probe phi variables cannot detect the violation.

**Experiment 3: Wrong intermediate variable.**  $\mathcal{M}$  implements the chain  $x \rightarrow m = 2x \rightarrow z = m + 3$ . The valid  $\mathcal{E}$  matches this exactly. The invalid  $\mathcal{E}$  posits  $m = 3x$  with a compensating downstream equation ( $z = \frac{2}{3}m + 3$ ), so that  $z = 2x + 3$  in both cases. The intermediate  $m$  is wrong, but the output is identical under observation.

**Experiment 4: Spurious mediator.**  $\mathcal{M}$  is fork-structured:  $x \rightarrow y$  and  $x \rightarrow z$  independently, so  $z$  does not depend on  $y$ . The invalid  $\mathcal{E}$  incorrectly posits the chain  $x \rightarrow y \rightarrow z$ . Specifically,  $y = 2x + 1$  and  $z = 3x + 2$  in  $\mathcal{M}$ ; the invalid chain  $\mathcal{E}$  posits  $z = 1.5y + 0.5$ , which reproduces  $z = 3x + 2$  observationally but fails under interventions on  $y$ .

**Experiment 5: Unreachable states.** Both  $\mathcal{E}$  and  $\mathcal{M}$  share the chain  $X \rightarrow M \rightarrow Z$  with  $M \in \{0, 1, 2, 3\}$ . The dynamics of  $\mathcal{M}$  only ever produce  $M \in \{0, 2\}$  from any input  $X \in \{0, 1\}$ . The valid  $\mathcal{E}$  uses the threshold  $M \geq 2$ ; the invalid  $\mathcal{E}$  uses  $M \geq 1$ , so they disagree only at  $M = 1$ , which is never reached from natural inputs. In micro-space,  $\mathcal{M}$  uses a threshold of 1.5 to cleanly separate the  $M \in \{0, 1\}$  subspaces from  $\{2, 3\}$  subspaces.

**Experiment 6: Wrong causal direction.**  $\mathcal{M}$  implements the chain  $x \rightarrow y \rightarrow z$  (so  $z$  depends on  $y$ ). The invalid  $\mathcal{E}$  posits the fork  $x \rightarrow y, x \rightarrow z$  (treating  $x$  as the common cause of both, ignoring  $y \rightarrow z$ ). Specifically,  $y = 2x + 1$  and  $z = 1.5y + 0.5$  in  $\mathcal{M}$ ; the invalid fork  $\mathcal{E}$  posits  $z = 3x + 2$ , reproducing  $z = 3x + 2$  observationally but failing under interventions on  $y$ .

## D Benchmark Systems

For each system, we describe the high-level model  $\mathcal{E}$ , the low-level model  $\mathcal{M}$ , the abstraction map  $\tau$ , the invalid conditions used to test discrimination, and key implementation details. All metrics are evaluated on a fixed number  $N$  of samples and averaged over  $n_{\text{runs}}$ . Their value, given below, were chosen to keep computational costs similar across systems.



## 1073 D.1 Logic Circuit

1074 A 2-bit ripple-carry adder evaluated at the wire level ( $\mathcal{M}$ ) and the gate level ( $\mathcal{E}$ ). This is the simplest  
1075 discrete-to-discrete abstraction in the benchmark.

- 1076 •  $\mathcal{E}$ : An SCM with integer-valued variables: two 2-bit operands ( $A, B \in \{0, \dots, 3\}$ ), a 1-bit carry-in,  
1077 a 1-bit internal carry, a 2-bit sum output, and a 1-bit carry-out. Each variable is a deterministic  
1078 Boolean arithmetic function of its parents.
- 1079 •  $\mathcal{M}$ : A Boolean netlist simulator over individual wire states. Each wire carries a float value in  
1080  $\{0.0, 1.0\}$ , and gates are simulated by iterating until stable. Interventions force specific wire values  
1081 before propagation.
- 1082 • **Abstraction map:** The coarse-graining map groups pairs of 1-bit wires or single wires into  
1083 variables of  $\mathcal{E}$ . The value map maps each integer label to the Cartesian product of unit-width  
1084 subspaces:  $(-0.1, 0.1)$  for bit-0 and  $(0.9, 1.1)$  for bit-1. Internal gate wires are declared as internal  
1085 variables and excluded from the phi set. Because every wire is either explicitly coarse-grained or  
1086 declared internal, the phi set is empty.
- 1087 • **Invalid conditions:**  
1088 (i) *Fail*:  $\mathcal{E}$  replaces XOR with OR in the sum logic.  
1089 (ii) *Inverted internal*:  $\mathcal{E}$  inverts `Internal_Carries` but compensates downstream, so the final  
1090 output is correct, but the intermediate is wrong.  
1091 (iii) *Noise*: Gaussian noise ( $\sigma = 0.15$ ) is added to wire voltages.
- 1092 • **Implementation details:** Metrics are evaluated over  $N = 1000$  samples and averaged over 100  
1093 runs.

## 1094 D.2 Transistor Circuit

1095 A CMOS half-adder (sum + carry) simulated at the SPICE transistor level ( $\mathcal{M}$ ) and the Boolean gate  
1096 level ( $\mathcal{E}$ ). This tests a continuous-to-discrete abstraction in which thresholding continuous voltages  
1097 yields Boolean gate values.

- 1098 •  $\mathcal{E}$ : A Boolean gate network with variables  $\{a, b, \text{sum}, \text{carry}\} \in \{0, 1\}$ , implementing XOR for  
1099 sum and AND for carry.
- 1100 •  $\mathcal{M}$ : A PySpice SPICE simulation of the CMOS half-adder topology. The topology uses four  
1101 NAND gates and one inverter to implement the half-adder, and produces continuous node voltages  
1102 for all circuit nodes (inputs, outputs, internal NAND wires, transistor junctions, and power rail).  
1103 Internal nodes are declared as internal variables and excluded from the phi set.
- 1104 • **Abstraction map:** The coarse-graining map associates  $\{a, b, \text{sum}, \text{carry}\}$  nodes to the correspond-  
1105 ing variables of  $\mathcal{E}$ . The value map maps label 0 to voltages in  $(-0.5, 0.5)$  V and label 1 to  $(4.5, 5.5)$   
1106 V. Voltages outside these ranges are unmapped.
- 1107 • **Invalid conditions:**  
1108 (i) *Fail*:  $\mathcal{E}$  uses OR instead of XOR for sum.  
1109 (ii) *Noise*: Gaussian noise ( $\sigma = 1.5$  V) is added to circuit outputs.
- 1110 • **Implementation details:** Metrics are evaluated over  $N = 250$  samples and averaged over 50 runs.

## 1111 D.3 Gas Simulation

1112 An  $N$ -particle Lennard-Jones fluid ( $\mathcal{M}$ ) evaluated against the ideal gas law or Van der Waals equation  
1113 of state ( $\mathcal{E}$ ). This is an extreme example of global aggregation: every variable of  $\mathcal{E}$  aggregates over  
1114 all  $N$  particles.

- 1115 •  $\mathcal{E}$ : The ideal gas law  $PV = NT$  (reduced units) or the Van der Waals equation  $(P + a\rho^2)(1/\rho -$   
1116  $b) = T$ , with causal variables  $\{P, V, T\}$ . Unless specified, the causal graph is reduced to the acyclic  
1117  $(V, T) \rightarrow P$  forward macroscopic mapping (NVT ensemble).
- 1118 •  $\mathcal{M}$ : An  $N$ -body molecular dynamics simulation using the 12-6 Lennard-Jones potential ( $N = 128$   
1119 particles), velocity-Verlet integration, Langevin thermostat, and Berendsen barostat. Long-range  
1120 tail corrections are applied analytically to the pressure virial to account for truncation of the

1121 potential at cutoff  $r_c = 3.0\sigma$  in reduced units.  $\mathcal{M}$  outputs the measured  $P, T, V$  alongside particle  
 1122 positions and velocities. An energy minimization step (steepest descent, up to 500 iterations) is  
 1123 applied after lattice initialization to remove particle overlaps before dynamics begin.

- 1124 • **Abstraction map:** The coarse-graining map is the identity on  $\{P, V, T\}$  (these are directly com-  
 1125 puted macroscopic observables returned by the simulator:  $T$  via kinetic energy,  $P$  via the virial  
 1126 theorem,  $V = \text{BoxLength}^3$ ). The value map is the identity.
- 1127 • **Invalid conditions:**  
 1128 (i) *Wrong  $\alpha$ :* the ideal gas  $\mathcal{E}$  uses an incorrect temperature exponent.  
 1129 (ii) *High density:* particle density is raised outside the ideal gas regime.  
 1130 (iii) *Low temperature:* temperature is reduced to a regime where the ideal gas law breaks down.
- 1131 • **Implementation details:** Van der Waals parameters  $a, b$  are calibrated against LJ fluid isotherms  
 1132 at the Boyle temperature ( $T^* \approx 3.42$ ). Metrics are evaluated over  $N = 10$  samples and averaged  
 1133 over 20 runs; a custom sampler initializes lattice configurations and hands equilibration to  $\mathcal{M}$ .

#### 1134 D.4 Predator-Prey Dynamics

1135 A spatially-explicit agent-based model of predator-prey dynamics ( $\mathcal{M}$ ) evaluated against the Lotka-  
 1136 Volterra ODE system ( $\mathcal{E}$ ). The abstraction is a continuous identity mapping over total population  
 1137 counts:  $\mathcal{M}$  outputs aggregate agent counts, which are the macro-variables.

- 1138 •  $\mathcal{E}$ : The Lotka-Volterra equations, with parameters  $\{\alpha, \beta, \delta, \gamma\}$  calibrated against the ideal (non-  
 1139 spatial, deterministic) ABM. The LV equations are integrated via the forward Euler method over  
 1140 50 steps, matching the ABM's timestep count.
- 1141 •  $\mathcal{M}$ : An agent-based model (ABM) of predator-prey dynamics in which individual prey and  
 1142 predators are tracked. Each timestep, births, predation events, and deaths are computed from  
 1143 per-capita probabilities. The simplest configuration (non-spatial, deterministic rates, no aging)  
 1144 approximates the LV equations and is used for calibration.
- 1145 • **Abstraction map:** The coarse-graining map is the identity on  
 1146  $\{\text{prey}_t, \text{predator}_t, \text{final\_populations}\}$ . The value map is the identity.
- 1147 • **Invalid conditions:** (i) *Wrong  $\alpha$ :* prey reproduction rate in  $\mathcal{E}$  is 3 times higher than in  $\mathcal{M}$ .  
 1148 (ii) *Spatial:*  $\mathcal{M}$  uses spatial dynamics on a  $20 \times 20$  grid, where agents move to and interact with  
 1149 neighboring cells.  
 1150 (iii) *Stochastic:*  $\mathcal{M}$  uses stochastic reproduction.  
 1151 (iv) *Aging:* agents have finite lifespans.  
 1152 (v) *Noise:* Gaussian noise ( $\sigma = 5$ ) on output populations.  
 1153 (vi) *Complex:* all three  $\mathcal{M}$  extensions simultaneously active.
- 1154 • **Implementation details:** Metrics are evaluated over  $N = 50$  samples and averaged over 100 runs.  
 1155 LV parameters are calibrated via grid search.

#### 1156 D.5 Heat Equation (1D)

1157 Brownian particle diffusion ( $\mathcal{M}$ ) is abstracted to a finite-difference heat equation ( $\mathcal{E}$ ). This tests a  
 1158 spatial aggregation abstraction.

- 1159 •  $\mathcal{E}$ : A finite-difference heat equation solver on a uniform grid of  $K$  bins, parameterized by the  
 1160 diffusion coefficient  $D$ . The finite-difference solver uses Neumann boundary conditions (zero flux  
 1161 at both walls), consistent with the reflective particle boundaries in  $\mathcal{M}$ .
- 1162 •  $\mathcal{M}$ : A system of  $N = 1000$  Brownian particles diffusing in a 1D box  $[0, L]$  with reflective  
 1163 boundaries over 200 time steps.
- 1164 • **Abstraction map:** The value map bins particle positions into a normalized density histogram (one  
 1165 bin per cell in  $\mathcal{E}$ ), and grounding samples particle positions from that histogram as a probability  
 1166 density function.
- 1167 • **Invalid conditions:**  
 1168 (i) *Fail:* the diffusion coefficient in  $\mathcal{E}$  is set to a wrong value of zero  
 1169 (ii) *Noise:* Gaussian noise ( $\sigma = 0.05$ ) on the outputs of  $\mathcal{M}$ .

1170 • **Implementation details:** Metrics are evaluated over  $N = 30$  samples and averaged over 50 runs.  
 1171  $K = 50$  bins. The diffusion coefficient  $D = 0.1$  is set to a known value matching the particle  
 1172 simulation’s diffusion constant. Initial conditions are sampled as random Gaussian profiles (mean  
 1173  $\sim U(0.3, 0.7)$ , width  $\sim U(0.05, 0.15)$ ), grounded to particle positions and re-abstracted to obtain  
 1174  $\mathcal{E}$ -compatible inputs.

## 1175 D.6 Heat Equation (2D)

1176 A phonon lattice model ( $\mathcal{M}$ ) is abstracted to a 2D heat PDE ( $\mathcal{E}$ ). This tests a spatial aggregation  
 1177 abstraction.

- 1178 •  $\mathcal{E}$ : A 2D heat PDE solver with Neumann boundaries and a calibrated diffusion coefficient  $\alpha$ .
- 1179 •  $\mathcal{M}$ : A  $16 \times 16$  phonon lattice model with harmonic springs, phonon scattering (rate  $p_{\text{scatter}} = 0.2$ ),  
 1180 and  $N_{\text{avg}} = 10$  simulation averages per sample.
- 1181 • **Abstraction map:** The value map abstracts the kinetic energy map via Gaussian smoothing  
 1182 ( $\sigma = 0.05 \times \text{GridSize}$ ) and normalization; grounding uses a parameterized Gaussian source. The  
 1183 source parameters (source\_x, source\_y, source\_E) use identity grounding.
- 1184 • **Invalid conditions:**  
 1185 (i) *Fail*: the diffusion coefficient in  $\mathcal{E}$  is set to a wrong value of  $10 \times$  the calibrated value  
 1186 (ii) *Noise*: Gaussian noise ( $\sigma = 50$ ) on the output of  $\mathcal{M}$ .
- 1187 • **Implementation details:** Metrics are evaluated over  $N = 5$  samples and averaged over 20 runs.  
 1188 The diffusion coefficient  $\alpha$  is calibrated via scalar optimization.

## 1189 D.7 Ising Model

1190 A hybrid molecular-dynamics and Monte-Carlo spin simulation ( $\mathcal{M}$ ) evaluated against a rigid-lattice  
 1191 Ising model ( $\mathcal{E}$ ). Both models operate over continuous thermodynamic variables; the internal discrete  
 1192 spins of  $\mathcal{M}$  are not exposed as micro-variables.

- 1193 •  $\mathcal{E}$ : A rigid  $L \times L$  Ising model simulated via Metropolis-Hastings MCMC with coupling constant  
 1194  $J$ . The causal variables are  $\{\text{Temperature}, \text{ExternalField}, \text{PredictedMagnetization}\}$ , all  
 1195 continuous.
- 1196 •  $\mathcal{M}$ : A hybrid molecular-dynamics/Monte-Carlo simulator in which atomic positions evolve via  
 1197 Newtonian dynamics when vibrational temperature  $> 0$ , and spins are updated via Metropolis  
 1198 sweeps with distance-dependent coupling  $J_{\text{eff}} = J_0 \exp(-\lambda|r - r_0|)$ .
- 1199 • **Abstraction map:** The coarse-graining map is the identity on all three continuous variables. The  
 1200 value map is the identity.
- 1201 • **Invalid conditions:**  
 1202 (i) *Fail*: the coupling constant in  $\mathcal{E}$  is set to  $J = 2$  (double the true value).  
 1203 (ii) *Noise*: Gaussian noise ( $\sigma = 0.05$ ) on the magnetization output of  $\mathcal{M}$ .
- 1204 • **Implementation details:**  $L = 8$ . Metrics are evaluated over  $N = 30$  samples and averaged over  
 1205 150 runs. Sampling is performed over  $[\text{Temperature}, \text{ExternalField}]$ . The explicit Euler PDE  
 1206 solver uses adaptive sub-stepping: when the stability criterion  $\alpha \Delta t / \Delta x^2 > 0.2$  is violated, the  
 1207 time step is subdivided accordingly.

## 1208 D.8 Tracr Transformer

1209 A transformer mechanistically compiled from a RASP sort-rank program ( $\mathcal{M}$ ) evaluated against the  
 1210 symbolic program itself ( $\mathcal{E}$ ). In this system,  $\mathcal{M}$  is a neural network.

- 1211 •  $\mathcal{E}$ : A symbolic sort-rank program over sequences of length  $L = 3$ :  $\text{rank}_i = |\{j \neq i : \text{token}_j <$   
 1212  $\text{token}_i\}|$ . Variables:  $\text{token}_i \in \{1, \dots, 5\}$  and  $\text{rank}_i \in \{0, \dots, 4\}$ .
- 1213 •  $\mathcal{M}$ : The Tracr-compiled JAX transformer, exposing two multi-dimensional micro-variables:  
 1214  $\text{tokens} \in \mathbb{R}^L$  (float encoding of token indices) and  $\text{ranks} \in \mathbb{R}^L$  (float encoding of rank values).

- 1215 • **Abstraction map:** The coarse-graining map slices  $\text{token}_i \rightarrow \text{tokens}[i]$  and  $\text{rank}_i \rightarrow \text{ranks}[i]$ .  
 1216 The value map uses  $[(v-1)-0.5, (v-1)+0.5]$  for token label  $v$  (tokens are 0-indexed internally:  
 1217 label  $v = 1 \rightarrow 0.0, v = 5 \rightarrow 4.0$ ) and  $[r-0.5, r+0.5]$  for rank label  $r$ . Tracr guarantees lossless  
 1218 round-trip encoding: grounding and abstraction are exact inverses for integer labels.
- 1219 • **Invalid conditions:**  
 1220 (i) *Fail*:  $\mathcal{E}$  uses  $>$  instead of  $<$  in the rank comparison (reversed ranks).  
 1221 (ii) *Noise*: Gaussian noise ( $\sigma = 0.3$ ) on the residual stream before read-out.
- 1222 • **Implementation details:** Metrics are evaluated over  $N = 100$  samples and 150 runs. The sequence  
 1223 length is  $L = 3$ , the vocabulary  $\{1, \dots, 5\}$ . Inputs are formed as  $[\text{BOS}] + [\text{vocab\_labels}]$ ; the  
 1224 BOS output position is discarded from the decoded result.

## 1225 D.9 Gene Regulatory Network (GRN)

1226 The segment polarity gene regulatory network of [Sánchez et al. \[2008\]](#) ( $\mathcal{M}$ ) evaluated against a  
 1227 simple two-variable Boolean causal rule ( $\mathcal{E}$ ).  $\mathcal{M}$  is a synchronous multi-valued Boolean network (not  
 1228 a continuous ODE model).

- 1229 •  $\mathcal{E}$ : A two-variable Boolean SCM:  $\text{wg\_src} \in \{0, 1\}$  (Wg signal active?)  $\rightarrow \text{fz\_tgt} \in \{0, 1\}$   
 1230 (target Fz on?). Rule:  $\text{fz\_tgt} = \text{wg\_src}$ .
- 1231 • **Low-level model:** One synchronous update step of the 6-cell segment polarity network.  
 1232 Variables are multi-valued Boolean (e.g.  $\text{Wg} \in \{0, 1, 2\}$ ), and updates follow GINsim log-  
 1233 ical parameter semantics. Micro-variables exposed:  $\text{wg\_c2}$  (Wg level in cell 2, input) and  
 1234  $\text{fz\_c1}, \text{fz\_c3}, \text{fz\_c4}, \text{fz\_c5}$  (Fz in various cells, outputs).
- 1235 • **Abstraction map:** The coarse-graining map links  $[\text{wg\_c2}] \rightarrow \text{wg\_src}$  and  $[\text{fz\_c1}, \text{fz\_c3}] \rightarrow$   
 1236  $\text{fz\_tgt}$  (true neighbors of cell 2). The value map ( $\text{WgFzValueMap}$ ) thresholds:  $\text{wg\_src} = 1$  iff  
 1237  $\text{wg\_c2} \geq 1.9$  (effectively  $=2 = 2 = 2$  for integer-valued states);  $\text{fz\_tgt} = 1$  iff any target Fz  $> 0.5$ .
- 1238 • **Invalid conditions:**  
 1239 (i) *Wrong map*: the coarse-graining links  $[\text{fz\_c4}, \text{fz\_c5}]$  (non-neighbors) instead of  
 1240  $[\text{fz\_c1}, \text{fz\_c3}]$ .  
 1241 (ii) *Wrong  $\mathcal{E}$* : the causal rule is reversed ( $\text{fz\_tgt} = 1 - \text{wg\_src}$ ).  
 1242 (iii) *Noise*: Gaussian noise ( $\sigma = 0.4$ ) on Fz outputs.
- 1243 • **Implementation details:** Metrics are evaluated over  $N = 200$  samples and averaged over 200  
 1244 runs. Sampling is performed over  $[\text{wg\_src}]$ .

## 1245 D.10 MOS 6502 CPU

1246 The MOS 6502 CPU, modeled at three discrete abstraction levels: the Visual6502 digital netlist, the  
 1247 manually extracted gate-level implementation, and a corresponding ISA-level simulation. All three  
 1248 levels are discrete: the transistor level uses binary switch segments, the gate level uses Boolean gate  
 1249 outputs, and the ISA level uses integer register and flag values (0–255). We evaluate abstractions  
 1250 between those three layers independently.

- 1251 • **Highest-level model (ISA):** The ISA semantics of selected 6502 instructions, expressed as a causal  
 1252 SCM over register variables (accumulator, index registers, status flags) implemented by the ISA  
 1253 simulator ([ivop/fake6502](#), commit b52676f).
- 1254 • **Intermediate-level model (gate):** A gate-level simulator ([ivop/break6502](#), commit 922af64),  
 1255 which implements the decoder PLA and datapath at the Boolean gate level.
- 1256 • **Lowest-level model (transistor):** A transistor-level digital logic simulator  
 1257 ([mist64/perfect6502](#), commit 6c1f1a5), extracted directly from the 6502 CPU via the  
 1258 [Visual6502](#) project. This implements a segment model in which each transistor is a binary switch.
- 1259 • **Abstraction maps:** All simulators share the same interface: integer register values 0–255 per  
 1260 variable. The coarse-graining map and value maps are both effectively identity on these integer  
 1261 values. All simulators mask the processor status register P with 0xCF (clearing bits 4 and 5) before  
 1262 comparison, since bit 4 (the B-flag) is only meaningful during hardware interrupts and bit 5 is  
 1263 hardwired high; without this mask, all status-affecting instructions would appear to disagree.

- 1264 • **Invalid conditions:**
- 1265 (i) *Broken gate*  $\leftrightarrow$  *ISA*: accumulator bit 7 is stuck at 0 in  $\mathcal{M}$ .
- 1266 (ii) *Broken transistor*  $\leftrightarrow$  *gate*: same fault applied at the transistor level.
- 1267 • **Implementation details:** The abstraction is evaluated on a reduced set of single-instruction tests
- 1268 covering implied and immediate addressing modes. Metrics are evaluated over  $N = 200$  samples
- 1269 and averaged over 100 runs (per condition).

## 1270 E Baseline Metric Implementation Details

1271 We document design choices, normalization functions, and known limitations for the baseline metrics.  
 1272 All metrics receive the same sampler,  $n$  samples, and output variable list as the CAE variants.

1273 **Observational metrics (MSE, RMSE, NMSE,  $R^2$ ,  $L^2$ , JSD, KL, MMD, HSIC, VarDecomp).**  
 1274 These are evaluated via a task wrapper that restricts interventions to root variables only and compares  
 1275 outputs at the designated output nodes.

1276 *JSD/KL*: Uses adaptive histogram binning into  $\lceil \log_2 n + 1 \rceil$  bins, capped at 20. KL adds Laplace  
 1277 smoothing ( $\varepsilon = 10^{-10}$ ). When samples collapse to a point mass, JSD returns 0.

1278 *MMD*: Uses an RBF kernel and bandwidth  $\sigma = \sqrt{\text{median}(\|x_i - x_j\|^2)/2}$  from the pooled sample  
 1279 (median heuristic). The unbiased  $\widehat{\text{MMD}}^2$  estimator is clipped to  $[0, \infty)$ , then passed through  $\tanh$ .

1280 *HSIC*: Measures whether the residuals  $R = \mathcal{M}(u) - \mathcal{E}(u)$  are statistically independent of the  
 1281 outputs of  $\mathcal{E}$  (used as a proxy for the inputs  $u$ ). The centered kernel alignment estimator  $\widehat{\text{HSIC}} =$   
 1282  $\text{tr}(HK_X H \cdot HK_R H)/(n-1)^2$  is computed, where  $K_X$  is the RBF kernel matrix of the outputs of  $\mathcal{E}$   
 1283 and  $K_R$  is the RBF kernel matrix of residuals; both kernel matrices are centered by  $H = I - \mathbf{1}\mathbf{1}^\top/n$ .  
 1284 The estimator is normalized by the product of the Frobenius norms of  $HK_X H$  and  $HK_R H$ . Requires  
 1285  $\geq 20$  samples; returns NaN below this threshold.

1286  $R^2$ : Normalized as  $(1 - R^2)/2$ , clipped to  $[0, 1]$ .

1287 *VarDecomp*: Reports  $\text{Var}(y - \hat{y})/\text{Var}(y)$ , which is robust to constant offsets.

1288 **Temporal metrics (TrajMSE, DTW, Autocorr, Spectral, SINDy).** These are implemented for the  
 1289 predator-prey system in a way that rolls out both  $\mathcal{E}$  and  $\mathcal{M}$  from shared initial conditions.

1290 *DTW*: Standard dynamic programming with Euclidean local cost, normalized by the sum of sequence  
 1291 lengths  $|s_1| + |s_2|$ .

1292 *SINDy*: For each sampled trajectory, we extract consecutive macro-step pairs  $(X_t, X_{t+1})$  from the  
 1293 trajectory of  $\mathcal{M}$  (ABM), where each step corresponds to one full model epoch. The observed epoch  
 1294 change is  $\dot{X}_t = X_{t+1} - X_t$ . The predicted change in  $\mathcal{E}$  is  $\hat{X}_t = \nu(X_t) - X_t$ , where  $\nu(X_t)$  runs  
 1295 the LV equations for one full epoch starting from the actual ABM state  $X_t$ . Only steps where both  
 1296 species exceed 1 individual are scored. The normalized Frobenius residual  $\|\dot{X} - \hat{X}\|_F / \|\dot{X}\|_F$  is  
 1297 mapped through  $\tanh(\cdot)$ .

1298 **Sobol first-order indices:** The Saltelli estimator shares the same  $A$  and  $B$  sample matrices for  
 1299  $\mathcal{E}$  and  $\mathcal{M}$ , making indices directly comparable. The metric reports mean  $|S_i^\mathcal{E} - S_i^\mathcal{M}|$  over input  
 1300 variables.

1301 **Infidelity (two-model attribution adaptation):** For each abstract input variable  $i$ , a finite-  
 1302 difference perturbation of size  $\delta_i$  is applied to the abstract label:  $\delta_i = \max(\delta_{\text{frac}} \cdot |x_i|, \delta_{\text{min}})$   
 1303 for continuous labels (with  $\delta_{\text{frac}} = 0.1$ ,  $\delta_{\text{min}} = 10^{-3}$ ) and  $\delta_i = 1$  for discrete (integer) labels.  
 1304 Both  $\mathcal{E}$  and abstracted  $\mathcal{M}$  are evaluated at the original and perturbed inputs; per-input attributions  
 1305  $\phi_i = (\text{output}_{\text{perturbed}} - \text{output}_{\text{original}})/\delta_i$  are computed for each model and compared by squared  
 1306 difference, averaged over input variables and output dimensions. The metric is normalized via  $\tanh$ .  
 1307 For the 6502 CPU, only register inputs  $\{A, X, Y, S, P\}$  are perturbed; opcode and operand bytes  
 1308 are excluded because a perturbation of  $\delta = 1$  changes the instruction identity, producing spurious  
 1309 attribution disagreement even for correct abstractions.



1310 **Relational fidelity:** Pearson  $r$  per output dimension, averaged, mapped to  $(1 - r)/2 \in [0, 1]$ .

1311 **Mallow’s  $C_p$ :**  $p = 2|\mathcal{V}^\mathcal{E}|$  (slope + intercept per variable of  $\mathcal{E}$ );  $\hat{\sigma}^2$  is estimated as the mean variance  
1312 of the outputs of  $\mathcal{M}$  across output dimensions, serving as a reference noise floor. Normalized as  
1313  $\tanh(\max(0, C_p - p)/N)$ .

1314 **Structural deviation and causal sensitivity index:** Structural deviation perturbs each parameter  
1315 of  $\mathcal{E}$  by 1% relative and measures  $|\text{CAE}_{\uparrow\text{NF}}(\mathcal{E}_{\text{perturb}}) - \text{CAE}_{\uparrow\text{NF}}(\mathcal{E}_{\text{base}})|$ . Causal sensitivity zeros each  
1316 parameter instead, but measures the same quantity. Neither metric divides by the baseline  $\text{CAE}_{\uparrow\text{NF}}\text{score}$ .

1317 **IB vs. CIB Lagrangian:** IB uses passive co-occurrence statistics for both  $I(X; T)$  and  $I(T; Y)$ ;  
1318 a spurious correlate of  $Y$  in  $\mathcal{M}$  can inflate  $I(T; Y)$  even without causal influence. CIB replaces  
1319  $I(T; Y)$  with  $\text{CAE}_{\uparrow\text{NF}}(Y \mid \text{do}(T))$ , estimated as  $H(Y) - H_c(Y \mid \text{do}(T))$  where the conditional  
1320 entropy averages over 20 label bins under direct interventions. Both Lagrangians are normalized to  
1321  $[0, 1]$ : the IB result is shifted by  $\beta H(Y)$  and divided by  $H(X) + \beta H(Y)$ ; the CIB result is shifted  
1322 by  $\beta$  and divided by  $H(X) + \beta$ , where  $H(\cdot)$  denotes histogram entropy of the respective variable.

1323 **Complexity shift:** For each sample, both the original input  $x$  and a Gaussian-perturbed input  
1324  $x + \varepsilon$  ( $\varepsilon \sim \mathcal{N}(0, 0.01^2)$ ) are run through both models. The complexity shift for each model is  
1325  $(K(\text{output}_{x+\varepsilon}) - K(\text{output}_x))/K(\text{input})$ , where  $K(\cdot)$  denotes the zlib level-9 compressed byte  
1326 length. The metric reports the absolute difference between the mean complexity shifts of  $\mathcal{E}$  and  $\mathcal{M}$ .  
1327 The metric is sensitive to output dimensionality and should be interpreted as a coarse ordinal measure  
1328 only.

1329 **Symbion:** Exhaustively enumerates all root-variable label combinations; applicable only to systems  
1330 with finite discrete label spaces.

1331 **Macroscopic invariance vs. BCC:** Both compare abstracted outputs of  $\mathcal{M}$  for two micro-states in  
1332 the same abstract cell. For discrete value maps, both metrics sample two micro-states directly from  
1333 the subspace associated with each label; the difference is that macroscopic invariance iterates over  
1334 individual labels, while BCC iterates over the full Cartesian product of labels and sample pairs. For  
1335 continuous value maps, BCC draws a label via the sampler and calls the grounding function twice,  
1336 and skips pairs where grounding is the identity, as these are trivially consistent.

1337 **DCC:** Always uses hard label equality for the next-state comparison.

1338 **IIA:** The interchange copies raw micro-values of  $\mathcal{M}$  from the source run (not grounded preimages  
1339 of the source abstract label). This tests whether the actual internal micro-state of  $\mathcal{M}$  is causally  
1340 exchangeable, which is a stronger condition than checking any preimage of the source label. This  
1341 distinction vanishes for identity value maps.

1342 **Probing:** For discrete value maps (logic circuit, Tracr, GRN), abstract labels are integers and probe  
1343 predictions are rounded before scoring. For continuous identity value maps, labels are floats and the  
1344 probe is a standard linear regression. The probe is trained on  $n_{\text{train}}$  samples (default 200, overridden  
1345 per system) and evaluated on a held-out test set.

## 1346 E.1 Computation Costs

1347 All experiments were run on a consumer-grade laptop equipped with 16 GB of RAM and an Intel  
1348 i5-13600H CPU. The computation time required to produce experimental results is estimated as such:

- 1349 • Evaluation of baseline metrics on benchmark systems for valid and invalid abstraction:  
1350 approx. 100 hours, of which nearly half was allocated to system 3 (gas simulation)
- 1351 • Measurements of statistical power and convergence: approx. 50 hours, of which half for  
1352 system 3
- 1353 • Scaling laws for Tracr: Approx. 5 hours

In addition, functional tests were implemented and ran to check for correctness of the CPU implementations. Their total runtime is estimated at approx. 50 hours.

Finally, preliminary experiments, including debugging and prototyping, is estimated to have required twice as much compute as the production of final results (approx. 250 hours).

## F Case Study: Gas simulation (system 3)

We report additional experiments for the gas simulation system, which tests whether thermodynamic laws emerge as valid causal abstractions of a Lennard-Jones fluid simulation. Unless otherwise stated, experiments are performed at the Boyle temperature  $T_{\text{Boyle}} \approx 3.418$  [Glasser, 2002] and with a low density of  $\rho = 0.05$ , where inter-particles interactions are rare and ideal thermodynamic behavior is expected.

We evaluate two high-level models: the ideal gas law (IGL,  $PV \propto T$ ) and the Van der Waals equation (VdW,  $(P + a/V^2)(V - b) \propto T$ ). These models represent successive levels of physical refinement: the IGL assumes non-interacting point particles, while VdW is a historical correction that explicitly accounts for finite particle volume and pairwise attraction. Neither is exact, but they serve as a useful pair to test whether CAE can discriminate the quality of competing valid abstractions.

For each equation, we consider three causal directions corresponding to different thermodynamic ensembles: NVT ( $P \leftarrow f(V, T)$ ), NPT ( $V \leftarrow f(P, T)$ ), and a non-standard inverse problem PVT ( $T \leftarrow f(P, V)$ ), implemented via a proportional-gain temperature controller.

### F.0.1 Valid Abstractions

Table 2 confirms that both models achieve near-zero CAE across all three causal directions in the dilute limit. This demonstrates that at low density, the validity of the statistical mechanical mapping matters more than the precise choice of equation of state. The VdW equation is marginally more faithful across most conditions, reflecting its tighter fit to the LJ fluid.

| CAE                    | NVT ( $P \leftarrow f(V, T)$ ) | NPT ( $V \leftarrow f(P, T)$ ) | PVT ( $T \leftarrow f(P, V)$ ) |
|------------------------|--------------------------------|--------------------------------|--------------------------------|
| Ideal gas law          | 0.0231                         | 0.0397                         | 0.0278                         |
| Van der Waals equation | 0.0135                         | 0.0520                         | 0.0435                         |

Table 2: CAE<sub>↓</sub> for two high-level gas models across three causal directions (10 samples per evaluation).

### F.0.2 Destructive Perturbations

We next systematically violated the physical assumptions required for thermodynamic laws to emerge from particle dynamics. Results are shown in Figure 12 and Table 3.

**Density sweep ( $\rho$ ).** Increasing density from  $\rho = 0.05$  to  $\rho = 0.85$  in NVT mode introduces progressively stronger inter-particle interactions. The IGL error rises steadily, while VdW remains below 0.05 for significantly longer, exploiting its explicit correction terms. Both abstractions effectively fail (error  $\approx 1$ ) for  $\rho > 0.45$ .

**Temperature sweep ( $T$ ).** At low temperatures ( $T < 1.0$ ), particles cluster in attractive wells, violating the homogeneity assumptions of both laws, and both fail (error  $\approx 0.8$ ). As temperature increases, VdW recovers faster, reaching its minimum around  $T \approx 2.5$ . The IGL minimum falls in  $[3.25, 3.75]$ , consistent with the Boyle temperature where attractive and repulsive contributions cancel [Glasser, 2002]. Both models remain valid at high temperatures ( $T > 4.0$ , error  $< 0.1$ ) where kinetic energy dominates.

**Distorted exponent ( $\alpha$ ).** Replacing the IGL’s temperature dependence with  $P \propto T^\alpha$  and sweeping  $\alpha \in [0.5, 1.5]$  tests sensitivity to functional form. The CAE correctly identifies the physical ground truth, reaching near zero at  $\alpha = 1.0$  and increasing monotonically away from it: sub-linear deviations ( $\alpha < 1$ ) degrade error nearly linearly (reaching  $\approx 0.7$  at  $\alpha = 0.5$ ), while super-linear deviations are penalized more gradually ( $\approx 0.42$  at  $\alpha = 1.5$ ).

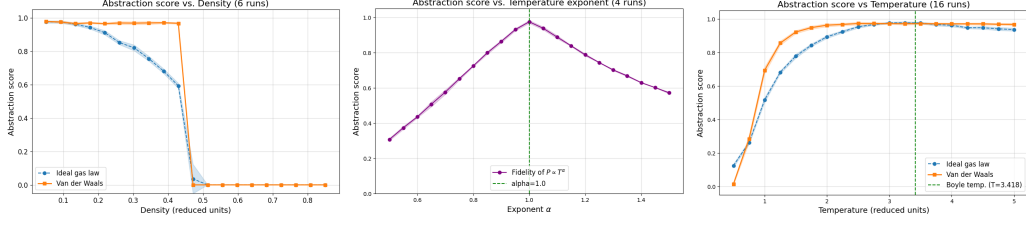


Figure 12: Average  $CAE_{\downarrow}$  for IGL and VdW under (a) increasing density, (b) varying temperature exponent  $\alpha$ , and (c) varying initial temperature  $T$ . All experiments in NVT mode, 10 samples per evaluation. Shaded areas denote  $\pm 1\sigma$ .

| $CAE_{\downarrow}$     | NVT ( $P \leftarrow f(V, T)$ ) | NPT ( $V \leftarrow f(P, T)$ ) | PVT ( $T \leftarrow f(P, V)$ ) |
|------------------------|--------------------------------|--------------------------------|--------------------------------|
| Ideal gas law          | 0.5920                         | 0.5432                         | 0.5497                         |
| Van der Waals equation | 0.5892                         | 0.5478                         | 0.5551                         |

Table 3:  $CAE_{\downarrow}$  for both models in non-equilibrium conditions (10 samples per evaluation).

1395 Together, these results illustrate that  $CAE_{\downarrow}$  behaves as a graded physical validity score: it correctly  
1396 validates both abstractions in their regime of applicability, distinguishes the more faithful VdW model  
1397 at intermediate densities and temperatures, and detects violations as soon as the underlying physical  
1398 assumptions break down.

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